

Mobility Friction and Gender Disparities in Expert Advisory Work: Evidence from the U.S. Congress

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Abstract

Expert advisory work is often episodic rather than an ongoing role, yet it typically still requires physical presence on site, often far from where the expert is based. This requirement can limit the value organizations capture from outside expertise, because individual mobility constraints fall unevenly across otherwise comparable experts. We test this using academic economists’ testimony before the U.S. Congress, constructing a novel dataset of the universe of economists who testified between 1969 and 2023, matched to a hearing-specific pool of at-risk candidates identified through the semantic similarity of their published research. We exploit the COVID-19 pandemic’s abrupt shift from in-person to virtual hearings as an exogenous change in mobility requirements. Conditional on individual characteristics and distance to Washington, women have modestly lower odds of testifying than comparable men, though the gap is sensitive to the sample period. Distance from Washington lowers the odds of testifying for all economists, but does so more steeply for women, producing a female advantage near the capital that reverses into a growing disadvantage as distance increases. The shift to virtual hearings increased women’s odds of testifying, with the largest gains among those located farthest from Washington. These results identify the requirement of physical presence, not the cost of travel, as a distinct and previously overlooked constraint on women’s access to expert advisory work. Work-from-anywhere arrangements, by removing this requirement for engagements that do not strictly need it, offer organizations a direct way to broaden the pool of experts they can draw on.

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1 Introduction

Experts provide organizations and policymakers with specialized knowledge through formal advisory appointments, scientific advisory board service, consulting engagements, and public testimony (Perkmann et al., 2021, 2013). Yet scientific expertise does not translate equally into advisory influence. Male academics are nearly twice as likely as comparably qualified women to serve on corporate scientific advisory boards (Ding et al., 2013), a pattern echoed in university-industry collaboration (Tartari and Salter, 2015), patenting (Ding et al., 2006), and academic entrepreneurship (Abreu and Grinevich, 2017). Beyond raising concerns about equity within academia, limiting advisory appointments to a narrower pool of experts implies a misallocation of talent (Hsieh et al., 2019), and shapes which problems receive attention and whose interests are served in organizational decision-making (Einiö et al., 2023; Koning et al., 2021; Nielsen et al., 2018). Existing accounts of this gap emphasize individual choices or gatekeepers’ preferences (e.g., Ding et al., 2013; Tartari and Salter, 2015), but neither considers a structural feature of advisory work itself. Unlike knowledge that travels through publications, exercising advisory influence typically requires physical presence in a small number of decision-making hubs. This requirement produces *geographic mobility frictions* (Choudhury, 2022) that are well documented to fall unevenly across individuals, in particular women. This paper asks to what extent the requirement of physical presence for episodic advisory work constrains women’s access to it, and how much its removal changes their participation relative to men.

Many forms of expert advisory work are geographically concentrated, requiring travel to hubs of corporate headquarters such as Silicon Valley, or governmental agencies such as Washington DC (Audretsch and Stephan, 1996). Prior research shows that geographic proximity facilitates knowledge transfer and collaboration, while reducing mobility frictions expands collaborative activity (Catalini et al., 2020; Ding et al., 2010; Gittelman, 2007; Katz, 1994). These frictions are unlikely

to be borne equally. Women continue to shoulder a disproportionate share of caregiving responsibilities, making travel more costly and contributing to sorting into jobs that offer greater work-life balance (Barbulescu and Bidwell, 2013), stronger preferences for temporal flexibility (Goldin, 2014), shorter commuting distances (Barbanchon et al., 2021), and lower ability to accommodate work requiring temporal availability outside standard working hours (Gagliardi et al., 2024). What is less clear is whether these constraints extend to episodic professional opportunities, where the time-bounded and infrequent nature of the commitment may make temporary substitution of caregiving responsibilities more feasible. We examine this possibility in a setting where such engagements are episodic rather than ongoing.

To test whether geographic mobility friction shapes access to advisory influence, and contributes to gender disparities in who exercises it, we study university researchers’ testimony before the U.S. Congress. We do so as this setting combines a consequential form of expert advisory work with unusual opportunities for observation and causal identification. Hearings provide a comprehensive public record of expert participation (Ban et al., 2023a) and underwent an abrupt transition from exclusively in-person to virtual hearings during the COVID-19 pandemic, an exogenous reduction in geographic mobility friction that lets us compare participation before and after the requirement of physical presence was removed.

This strategy, however, depends on knowing not only who testified, but who else could plausibly have done so. Studying selection into expert advisory work presents an important empirical challenge because the relevant risk set is rarely observed. While many forms of expert engagement record who ultimately participates, organizations rarely document the broader set of experts who were eligible or considered for participation. We address this challenge by developing a novel approach to approximating the risk set in settings where individuals leave a public record of their expertise, such as academic scientists through publications. For each congressional hearing, we construct a hearing-specific risk set using only information available at the time of the hearing, combining observable characteristics with the semantic similarity of scholars’ published research to identify academics with comparable expertise to the invited witness. This approach enables the study of selection into episodic expert advisory roles by recovering an otherwise unobservable risk set and is applicable to contexts in which expertise can be inferred from an observable knowledge base.

We construct a novel dataset capturing the universe of academic economists who testified before Congress between 1969 and 2023, comprising 3,662 testimonies by 1,326 unique individuals, and combine it with our hearing-specific risk sets allows us to estimate selection into congressional testimony conditional on being a qualified candidate. Three findings emerge. First, evidence of a gender gap in the odds of testifying is mixed. Women’s odds are approximately 10 percent lower than comparable men’s once academic qualifications and geographic distance to Washington D.C. are accounted for, but this gap disappears once the sample is restricted to hearings from 2000 onward. Second, geographic distance from Washington D.C. lowers the likelihood of appearing

as an expert witness for all academics, and does so more steeply for women. Among academics located closest to the capital, women are in fact *more* likely than comparable men to testify, an advantage that reverses into a significant disadvantage as distance grows. Finally, consistent with our theoretical argument, the shift from in-person to virtual hearing formats produced large gains for female scientists, particularly those located farthest from Washington, the group whose travel requirements were reduced the most by removing the in-person requirement. Together, these findings identify geographic mobility friction as an important supply-side mechanism underlying gender gaps in access to expert advisory work.

First, we contribute to the literature on geography and knowledge work, which has shown that geographic frictions shape collaboration, knowledge transfer, and innovation (e.g., Jaffe et al., 1992; Katz, 1994; Ding et al., 2010; Catalini et al., 2020). Existing evidence, however, paints an incomplete picture. While reducing communication costs disproportionately benefits women scientists' collaborations and productivity (Ding et al., 2010), reducing travel costs appears to benefit primarily men (Catalini et al., 2020). We help explain these findings by studying expert advisory work, a setting in which communication can occur remotely but participation has traditionally required physical presence. Our findings suggest that an important dimension of geographic friction is not the cost of travel alone but the organizational requirement for in-person participation, highlighting how the design of knowledge work shapes who can participate in it.

Second, we contribute to the academic engagement literature by advancing understanding of expert advisory work as an important yet relatively understudied form of engagement between academics and organizations (Perkmann et al., 2013, 2021). We document substantial gender disparities in participation in these influential advisory roles and identify geographic mobility frictions as an important structural mechanism underlying these disparities. We further show that communication technologies that eliminate need for physical presence can mitigate these frictions and increase the participation of women in expert advisory work. Together, these findings extend existing accounts of academic engagement by highlighting how the organization of advisory work shapes who participates in it.

Third, we contribute to the literature on geographic mobility and gender inequality in labor markets, which has shown that differences in commuting, relocation, and workplace flexibility shape occupational sorting, career progression, and earnings (Blackburn, 2010; Goldin, 2014; Mincer, 1978; Peterson, 1979; Sandell, 1977; Schoenbaum, 2017; Sorenson and Dahl, 2016; Widiss, 2012; Borghorst et al., 2024; Casado-Díaz et al., 2023; Barbanchon et al., 2021; He et al., 2021; Wiswall and Zafar, 2018). We extend this literature to episodic professional opportunities that require temporary travel rather than permanent or long-term relocation. Our findings show that geographic mobility frictions influence access to prestigious professional opportunities even when mobility demands are infrequent and short-lived, suggesting that the consequences of mobility constraints extend beyond employment choices to the allocation of influential advisory roles.

The rest of this manuscript is organized as follows. Section 2 outlines the background literature on gender disparities in academic engagement and geographic mobility, from which hypotheses are derived. In section 3 congressional hearings as an empirical context is described and in section 4 and section 5 our data, variables, and methods are outlined. In section 6 our predictions are tested on the universe of academic economists who have advised members of Congress via hearing testimony from 1969-2023 with a constructed at-risk population of intellectually close academics. Finally, implications, limitations, and future work are discussed in section 7 and section 8.

2 Gender, Academic Engagement, and Geographic Mobility Friction

This section develops our hypotheses from two streams of literature. The first documents persistent gender disparities in academics' engagement with non-academic spheres of society and attributes them to a mix of supply and demand side mechanisms. The second establishes that geographic mobility is costly and that women bear that cost disproportionately, resulting in gender gaps in labor market outcomes. Advisory work sits at the intersection, as it typically requires geographic mobility, and there is some documentation of gender disparities in participation. Yet, the literature on academic engagement has largely set aside the spatial dimension while the literature on mobility has scarcely considered *episodic* professional opportunities of this kind.

2.1 Academics as Experts

Organizations and policymakers routinely defer to experts when making decisions under uncertainty (Bonaccio and Dalal, 2006; Vaaler and McNamara, 2004), and expert judgment measurably shapes the choices of those who receive it (Kawamura and Vlaseros, 2017). Among the sources of external expertise, academic researchers occupy a distinctive position. As producers of specialized knowledge, university researchers translate and disseminate that knowledge beyond the academy, driving innovation (Bikard and Marx, 2020; Sorenson and Fleming, 2004), informing public policy, and helping address pressing social and environmental challenges (Azoulay and Jones, 2020).

Scientific knowledge reaches relevant audiences through two broad channels. On the one hand, academic research creates economic value directly, as discoveries are patented, licensed, and carried into commercial application, a process through which codified scientific knowledge becomes a durable source of productivity and growth (Jaffe, 1989; Mansfield, 1991). On the other hand, academics advise firms and public policymakers (Ding et al., 2013; Jacobs and Page, 2005), lending not only expertise but a form of authority and legitimacy grounded in the Mertonian norm of disinterestedness, the expectation that scientists advance claims without regard to their own stake

in the outcome (Merton, 1973). Networks of credentialed experts have long been shown to shape how governments define problems and choose among policies under uncertainty (Haas, 1992), and legislative bodies turn to academic experts to supply an evidence base that members often lack (Ban et al., 2023b). We might expect an organization’s search for expertise to yield advisors selected on merit alone. But both the costs of engaging and the processes that select who is invited results in uneven selection, giving reason to expect participation to concentrate among a distinct segment of the academic population instead.

Four decades of research on academics’ engagement with non-academic spheres of society has produced a fairly consistent picture. Academic scientists who participate in engagement outside academia tend to be productive in the field, senior, locally trained, commercially experienced, and male (Perkmann et al., 2013, 2021). That profile rests almost entirely on studies of the commercial application of STEM research, including collaboration with industry (Abramo et al., 2011; Tartari and Salter, 2015), patenting and licensing (Whittington and Smith-Doerr, 2005; Thursby and Thursby, 2005), and academic entrepreneurship (Lacetera, 2009; Marx and Hsu, 2022), in part because publications can be traced to observable downstream outcomes such as patents and firm performance (Kogan et al., 2017). Yet this evidence is largely descriptive, documenting who engages, not what drives the concentration, and because commercialization and advisory work differ in their incentive structure, whether patterns established for commercial engagement extend to government advising remains open, an activity that has attracted far less scholarly attention (Abreu and Grinevich, 2013; Bornmann, 2013; Lawson, 2022).

2.2 Gender gaps in Academic Engagement

Of the characteristics that predict academic engagement, gender has proven one of the most persistent. Women are consistently less likely than otherwise comparable male academics to participate in commercialization activities, even among highly accomplished faculty (Abreu and Grinevich, 2017; Ding et al., 2006; Murray and Graham, 2007; Tartari and Salter, 2015).

Scholars have offered explanations for this pattern that fall under two broad theoretical lenses: Supply and demand-side. Supply-side accounts attribute the gap to individual differences in *preferences and choices*. The largest single explanation is field sorting. Women are concentrated in disciplines and subfields with weaker traceable downstream commercial application, which accounts for much of the raw gap in patenting (Hunt et al., 2013). Second, women continue to shoulder the majority of domestic work and care-taking responsibilities (Kleven et al., 2019; Tartari and Salter, 2015), leaving them with fewer hours for professional responsibilities, relative to men. Because engagement is rarely rewarded in promotion and tenure decisions, it competes with publishing for time, and a binding time constraint should therefore fall hardest on this discretionary work. Finally, there is a gender gap in the willingness to express expert claims publicly. On panels of leading

economists, women are less likely than men to offer an opinion, less likely to state it strongly, and less confident in its accuracy, especially when they have background uncertainty (Sarsons and Xu, 2021; Sievertsen and Smith, 2025). A further reason may be that women who do testify can face hostile treatment from legislators, which may further dampen their willingness to engage (Bisbee et al., 2025). Common to these accounts is that the constraint is located in the academic herself, in the field she works in, the time available to her, or her willingness to engage.

Demand-side accounts centers on the mechanisms that limit women’s *opportunities* in the workplace. A reoccurring explanation in the literature rests in network differences Whittington (2018). In the context of science advisory boards, Ding et al. (2013) find that men reach advisory roles through broad networks spanning strong and distant ties, whereas women depend on strong direct ties to close collaborators. Because these insiders can vouch for a candidate’s qualifications, such ties help offset in-group favoritism among selectors, leaving women without them far less likely to be identified. Stuart and Ding (2006) finds that academic life scientists become commercially active largely through exposure to already active collaborators, so network position rather than individual attributes drives entry. Second, Ding et al. (2013) also find evidence of bias in commercial partners’ evaluation of women academics, as past leadership roles have a much stronger effect on the likelihood of serving on a science advisory board for women than men. The demand-side accounts operates before the academic makes any choice of her own, through the influence of networks and gatekeepers.

These two accounts are not independent. Murray and Graham (2007) demonstrate an interplay between supply and demand factors in commercial science. Women’s initial exclusion from commercial opportunities limits their socialization into the work of commercial science. Because their networks are correspondingly less often activated for commercial ends, they are less able to resolve ambiguities surrounding engagement in its early years. The predominance of men among early participants meanwhile leads the commercial role itself to be construed as male, leaving women to underappreciate their own commercial competences in comparison with their male colleagues. What presents as a supply side difference in willingness or skill may therefore be the accumulated residue of demand side exclusion.

Expert advisory work shares features with commercial engagement. It is discretionary work that competes with research for scarce time and is typically not formally rewarded in tenure decisions, it requires stating expert judgment publicly and under scrutiny, and selection to it runs through the networks and evaluations of those who select the experts. We therefore expect:

Hypothesis 1. *Female academic scientists are less likely than comparable male academic scientists to serve as an expert witness before the US Congress.*

While the explanations offered above provide important insights into the gender gap in academic engagement participation, they attend little to the geographic boundaries of the engagement activity. Expert advisory work is spatially concentrated, often requiring travel to where policymakers

and organizational decision-makers are based, and the burden of such travel is not borne equally. Prior work has established that advisory relationships are geographically structured (Audretsch and Stephan, 1996), but viewed this structure as a feature of advisory work itself rather than as a source of unequal access. The gender and engagement literature, in turn, has examined unequal access while largely setting its spatial dimension aside. We combine these strands of literature, by arguing that geographic mobility friction is an additional and largely overlooked supply-side constraint on women’s participation in expert advisory work.

2.3 Geographic Mobility Friction and Gender Gaps in Academic Engagement

Geographic mobility is a key channel through which knowledge workers build networks (Agrawal et al., 2008; Singh, 2005), transfer knowledge (Almeida and Kogut, 1999; Ganguli, 2015), and create value for organizations and individuals (Choudhury, 2022; Saxenian, 2005). Mobility matters precisely because knowledge is spatially bounded. Knowledge spillovers are geographically localized, concentrating within regions and decaying with distance from their source (Jaffe et al., 1992; Katz, 1994). This decay is not merely a correlate of where research clusters. Using the deaths of inventors in co-inventor pairs to isolate the interpersonal component of knowledge transfer, Balsmeier et al. (2026) demonstrate lower local citation rates for deceased versus still-living inventors. This spatial structure extends to advisory relationships between academics and the organizations they inform (Audretsch and Stephan, 1996).

Because so much of knowledge embedded in experts is tacit and best transmitted through direct contact, and because geographic distance raises the cost for an academic to participate in knowledge sharing, we expect geographic proximity to be conducive to knowledge transfer between academics and legislators on Capitol Hill. We hypothesize:

Hypothesis 2. *Geographic distance between an academic’s home institution and Washington D.C. is negatively associated with their odds of appearing as an expert witness.*

Yet mobility is not costless, as friction from regulatory, organizational, financial, and personal factors constrain it. Crucially, these constraints fall disproportionately on certain groups, so that the capacity to convert mobility into professional advantage is itself unequally distributed (Choudhury, 2022). Ample evidence documents that women are, on average, less geographically mobile than men, and therefore more constrained wherever labor market participation demands spatial displacement. That constraint is apparent in the jobs they select, which offer shorter commutes (Borghorst et al., 2024; Casado-Díaz et al., 2023; Barbanchon et al., 2021) and greater work life balance (Barbulescu and Bidwell, 2013), a pattern that extends to self employment (Rosenthal and Strange, 2012). These patterns emerge well before careers begin. In educational choices, women prove less willing to study abroad or pursue geographically distant degrees (Farré and Ortega, 2024). It is even visible in everyday movement, where women visit fewer distinct locations and concentrate their movement

more narrowly than men (Gauvin et al., 2020). Geographic mobility constraints lead women to trade steeper wage growth for proximity and flexibility, accounting for a substantial share of the observed gender wage gap (Petrongolo and Ronchi, 2020). One might expect this pattern to weaken in academia, a profession that grants unusual autonomy over where and when work is performed. Yet academic work still routinely requires relocation, as well as travel, to conferences, collaborations, and invited engagements, and the same gender gap in mobility persists in this setting (Zhao et al., 2023).

One mechanism behind women’s lower mobility is caregiving roles, operating both at the household and the individual level. Mobility decisions are typically made jointly in households, and couples tend to relocate toward wherever the household’s expected return is highest. Because men more often occupy the higher earning position, a calculus that systematically favors the man’s career (Mincer, 1978; Blackburn, 2010). Sorenson and Dahl (2016) show that the relocation bias is neither incidental nor small, accounting for a substantial share of the gender wage gap, with the effect sharpest among couples with young children and a higher earning male partner. The asymmetry persists even after the relocation, where woman’s earnings typically fail to recover (Sandell, 1977; Schoenbaum, 2017). Underlying these relocation asymmetries is the unequal distribution of caregiving itself. Children are the principal driver of divergence in women’s labor supply (Goldin, 2014), and men and women follow parallel trajectories until parenthood, after which they separate sharply and durably (Cairo et al., 2026; Kleven et al., 2019).

The mobility frictions documented above were studied almost entirely in the contexts of permanent relocation between geographic boundaries, or recurring daily mobility (e.g. commuting). Episodic professional opportunities, like expert advisory work, often requires physical presence that is neither permanent nor continuous, but intermittent, binding, and layered onto the individual’s job. The further an academic’s institution lies from the site of engagement, the higher the cost of the physical presence it demands. Testifying becomes correspondingly less attractive relative to declining the invitation.

As geographic mobility costs tend to affect women more than men we expect the following moderating effect:

Hypothesis 3. *The negative association between geographic distance to Washington D.C. and the odds of appearing as an expert witness is stronger for female academics than for male academics.*

If mobility friction is what binds, then the constraint is not a feature of the work itself but a property of how the work is organized, and is therefore amenable to organization design (Arntz et al., 2022). Choudhury (2022) argues that provisioning geographic flexibility, through work-from-home and work-from-anywhere arrangements, allows individuals and organizations to capture the returns to mobility while mitigating its frictions. Organization’s shift from in-person to virtual hearings during the COVID-19 pandemic is one such arrangement, and could potentially weaken

the link between geographic distance and an academic’s odds of appearing as a witness.

Whether we would expect virtual arrangements to benefit women depends on the mechanism through which the virtual format relieves the geographic constraint. Prior evidence distinguishes two; the elimination of *travel requirements* and the reduction of *travel costs*. Technologies that *reduce the need to be present*, by letting scientists communicate and collaborate remotely, disproportionately benefit women scientists’ productivity and collaboration to a larger degree than men’s (Ding et al., 2010). The introduction of cheaper air travel, on the other hand, benefited only male scientists’ collaboration ties, not women’s (Catalini et al., 2020). Taken together, these findings suggest that the primary source of friction to women’s geographic mobility is *the requirement to travel* rather than the *monetary cost of that travel*. Virtual formats of knowledge exchange eliminate not only the monetary cost of traveling, but, more importantly, the requirement to travel itself.

If the mechanism behind Hypothesis 3 is mobility friction, rather than some other correlate of distance, removing the in-person requirement that governed hearings before the pandemic should encourage women’s participation, and do so disproportionately for geographically distant women. We therefore expect:

Hypothesis 4. *The shift from in-person to virtual hearing format increases the odds for female academics to appear as a witness to a larger degree than for men.*

Hypothesis 5. *The shift from in-person to virtual hearings increases the odds of a female academic to appear as an expert witness more for academics located farther from Washington D.C.*

3 Empirical Context

To test these hypotheses, we examine expert testimony before the U.S. Congress. This setting combines a consequential and highly visible form of episodic expert engagement with a geographic requirement, and importantly, one that changed abruptly and for reasons unrelated to any individual witness’s characteristics. Hearings are also unusually well documented, leaving a public record of who testified that we build on in constructing our sample. Below, we situate congressional testimony within academics’ broader engagement with policymakers, describe how testifying is organized from the perspective of both Congress and the witness, and outline the shift to virtual hearings that we use as a source of variation in mobility requirements.

3.1 Academic Expertise in Congressional Policymaking

Science informs decision making across both public policy and corporate domains, giving decision makers access to evidence they could not generate on their own (Prewitt et al., 2012). Policymakers

can source this expertise in different ways. One route runs through written artifacts, namely academic publications, which make explicit knowledge available to anyone willing to read the record. These form of knowledge sourcing is, to a limited extent, traceable through citations in policy documents (Distel et al., 2026). A second route runs through direct engagement with experts. Academics do so by serving on boards and committees (Lambright and Teich, 1978; Ding et al., 2013, e.g.), by taking political appointments (Hirschman and Berman, 2014, e.g.), or by testifying before legislative bodies (Ban et al., 2023a; Maher et al., 2020). Engagement of this kind gives policymakers access to tacit and synthesized knowledge that rarely fits into a single document. However, these forms of engagement are harder to study because they do not usually leave a public, systematic paper trail behind them. Testimony before Congress is an exception, producing a dated, detailed individual-level record of who engaged and when, one this study builds on directly.

From the organization’s perspective, the US Congress functions much like other organizations that must source knowledge they do not hold internally. The US Constitution vests Congress with the authority to write federal laws, control the national budget, and oversee the executive branch. Congress exercises these responsibility through committees, specialized subunits staffed by members and charged with developing expertise on the issues within their jurisdiction (Cushman, 2014). Much as managers turn to outside consultants or scientific advisory boards when internal capabilities fall short, members of Congress rely on outside witnesses, drawn from academia, industry, federal agencies, and interest groups, to inform decisions in areas where their own knowledge is limited (Ban et al., 2023a). The intensity of this reliance fluctuates, growing as the policy environment becomes more complex or uncertain (Ban et al., 2023a; Furnas et al., 2024; Jacobs and Page, 2005), a pattern that echoes how organizations more broadly intensify their search for outside expertise precisely when internal knowledge proves insufficient (Bonaccio and Dalal, 2006). Seen this way, congressional hearings are less of an idiosyncratic institutional ritual and more an instance of a general organizational problem, namely how to identify and select external experts, which situates our setting within a broader question of knowledge sourcing rather than treating it as unique to the congressional process.

From the witness’s side, testifying is unlike the other forms of engagement described above. It is a one-off event rather than an ongoing relationship, concentrated into a single appearance rather than spread across a term of service, yet it carries outsized visibility and impact relative to its brief duration. However, this opportunity comes at a cost. Testifying has historically required traveling to Washington, D.C. on a fixed date set by the committee, typically at the witness’s own expense, and preparing written and oral testimony under a compressed timeline. It can also carry risk, testimony on politically charged topics can draw public scrutiny or personal attacks in a way that a board seat or a consulting contract rarely does. Unlike service on a board, a formal research collaboration, a consulting arrangement, or appointment to a leadership role, testimony asks for no lasting commitment and comes with no strings attached, its entire cost and its entire exposure are borne in a single sitting.

That single sitting, however, has always taken the same physical form. Since the modern committee system took shape, hearings have meant a witness present in Washington, D.C., on a date fixed by the committee, giving researchers a stable baseline against which recent change can be assessed. That changed in 2020.

3.2 COVID-19 and the Introduction of Virtual Hearings

The change to congressional hearing format came from the 2020 COVID-19 global health emergency, not from a reform to the workings of Congress itself. On March 11, 2020, the WHO declared COVID-19 a global pandemic, and in the time that followed, organizations and institutions were forced to change their daily operations dramatically¹. Due to lockdowns and quarantine measures, communication was increasingly facilitated online. To address the resulting need for physical distancing, Congress adopted new rules and procedures to keep the legislative body running, chief among them a shift in hearing format from in-person to online or hybrid. The US Senate held its first virtual hearing on May 5th, 2020, and the US House of Representatives adopted rules allowing remote committee hearings on May 15th, 2020 (Jones, 2020).

This change allowed witnesses to give testimony from wherever they were, rather than from a committee room in Washington, D.C. Where testifying had always required travel to the Capitol, usually at the witness’s own expense, it now required only an internet connection. Once the risk of COVID-19 faded and health authorities changed their recommendations, Congress returned to the older format: since January 2023, hearings have been held mostly in person, though some hybrid formats remain in use. This shift from in-person to virtual, driven by public health policy rather than by any individual academic’s location or prior engagement with Congress, offers a source of variation in mobility requirements that we exploit throughout the remainder of the paper.

4 Data

This paper draws on multiple data sources to assemble a comprehensive dataset capturing the universe of academic economists who have testified before the U.S. Congress from 1969 to 2023, as well as a large at-risk group of plausible alternative witnesses. We build this dataset following a number of steps. We start by using data compiled by Maher et al. (2020), covering testimonies through 2016. We extend this data with a primary collection effort covering hearings from 2017 to 2023. We then construct an at-risk group for each witness appearance, drawing on the authors of papers semantically similar to the focal witness’ stock of publications prior to the hearing. Finally,

¹World Health Organization, “WHO Director-General’s opening remarks at the media briefing on COVID-19,” 11 March 2020, <https://www.who.int/news-room/speeches/item/who-director-general-s-opening-remarks-at-the-media-briefing-on-covid-19--11-march-2020>.

we supplement both the witness and at-risk data with publication output from OpenAlex, Altmetric indicators, and geographic distance to Washington DC from Google Maps, to build a career panel for each individual academic.

4.1 Economists as Witnesses in Congressional Hearing

The selection of economists as the focal population for this research is deliberate and methodologically driven. Economics is characterized by a remarkably stable and entrenched hierarchical structure, with minimal disruption to its institutional ecosystem over time (Fourcade et al., 2015). This stability is evident in the persistent dominance of a small set of elite academic institutions (Korom, 2020), the consistent prestige of top five journals (Heckman and Moktan, 2020), and the self-perpetuating networks of academic and professional recognition. The field operates through tightly-knit “elite clubs” – networks of institutions, research centers, and professional associations that exercise significant gatekeeping functions, making the boundaries and key actors relatively well-defined and traceable (Carrell et al., 2024; Freeman et al., 2024). Moreover, there is evidence to support the consideration of economics as a discipline with a strong policy orientation, with economists frequently serving as key advisors, consultants, and expert witnesses in policy formulation across various domains of governance (Hirschman and Berman, 2014; Maher et al., 2020).

From a methodological standpoint, there are two particular advantages of focusing on economists. First, our main source of variation, the COVID-19 pandemic, not only forced organizations to introduce technological tools that facilitated remote work, but also disrupted researchers’ established routines. A potential reduction in time devoted to research during the pandemic could affect our results, if it frees up the researcher’s time to devote to other tasks, such as testifying before Congress. Studying economists is useful since the bench sciences who rely on lab equipment to conduct their research, such as biochemistry and biology, reported a much larger disruption to their work compared to mathematics, statistics, computer science and economics (Myers et al., 2020). Second, economists are among the most frequently invited academic witnesses in Congress (Maher et al., 2020), which yields the data coverage needed to detect compositional shifts around the move to virtual hearings.

4.1.1 Congressional Hearings Data

To build our dataset of congressional witnesses, we began with data collected by Maher et al. (2020) which encompasses testimonies from social scientists appearing before U.S. congressional committees from 1946 to 2016, detailing information on witnesses, their affiliations, the committees involved, and the hearing topics. In total, the dataset comprises 23,016 testimonies by 7,295 unique social scientists². Economists constitute the majority of social scientists testifying: 15,050 testimonies,

²Including both university and non-university researchers.

or 65 percent of the witness appearances recorded by the researchers, were delivered by economists, representing 3,883 unique individuals.

We expanded this dataset with a primary data collection effort to include 2017-2023, as these years are critical to our empirical design. We did so by collecting hearing records from the US congressional archive GovInfo³ and extracting the witnesses using an API access to a Generative-AI tool developed by OpenAI (GPT-4o). In total, 7,381 hearing transcripts as PDF files were collected.

To extract the precise information about witnesses appearing in congressional hearings, a Python script was developed, which contained three steps. In the first step, each hearing file was parsed using keywords indicating the presence of witness information and truncated to limit the amount of tokens spent. In the second step, a prompt containing a small sample of the desired data schema was sent to GPT-4o⁴. For consistency, the extracted variables followed those in the Maher et al. (2020) dataset, with one addition; an indicator of whether a hearing was facilitated online or in-person, as this information is crucial for our identification strategy. The third and final step entailed the organization of the output data in a CSV format.

The extraction identified 27,582 witnesses (18,602 unique individuals) out of which 3,512 (2,855 unique individuals) were identified as academics. For 3,391 of these, GPT-4o also inferred a field (e.g., professor of management), and we manually reviewed each case to determine which fields fell within economics. For the remaining 121 witnesses, GPT-4o did not infer a field, so we identified it manually using information from their CV and university page. Removing non-economists reduced the sample to 942 economists (742 unique individuals).

Figure 1 places this new data collection in context by comparing the number of academic congress witnesses per year across data sources. Maher et al. (2020) track social scientists and economists in congressional testimony between 1946 and 2016⁵. Ban et al. (2023a) offer a more inclusive benchmark, covering all think-tank and university witnesses from 1960 to 2018, in which think-tank and university researchers made up 8.45 percent of the total witness pool. We include this series for reference only, as our analysis does not draw on it directly. Our own sample of economists, by contrast, is a subsample of Maher et al. (2020) restricted to those holding a university affiliation at the time of the hearing, and the red segment extends this series with the academic economists identified in our 2017-2023 collection. Read together, the four series show a broadly consistent pattern over time. The rise in witnesses around 1970 coincides with the Subcommittee Bill of Rights of 1974, which expanded subcommittees' autonomy and jurisdiction and encouraged committees to create more of them and take on a wider range of issues (Rohde and Rosenthal, 1974), while the decline visible from the 1990s onward is consistent with a 1995 reform that reduced the number of subcommittees and partly reversed these effects (Ban et al., 2023a).

³<https://www.govinfo.gov/app/collection/chrg/>.

⁴See prompt in appendix A.

⁵Not just academic economists, but also those working at think tanks, government agencies etc.

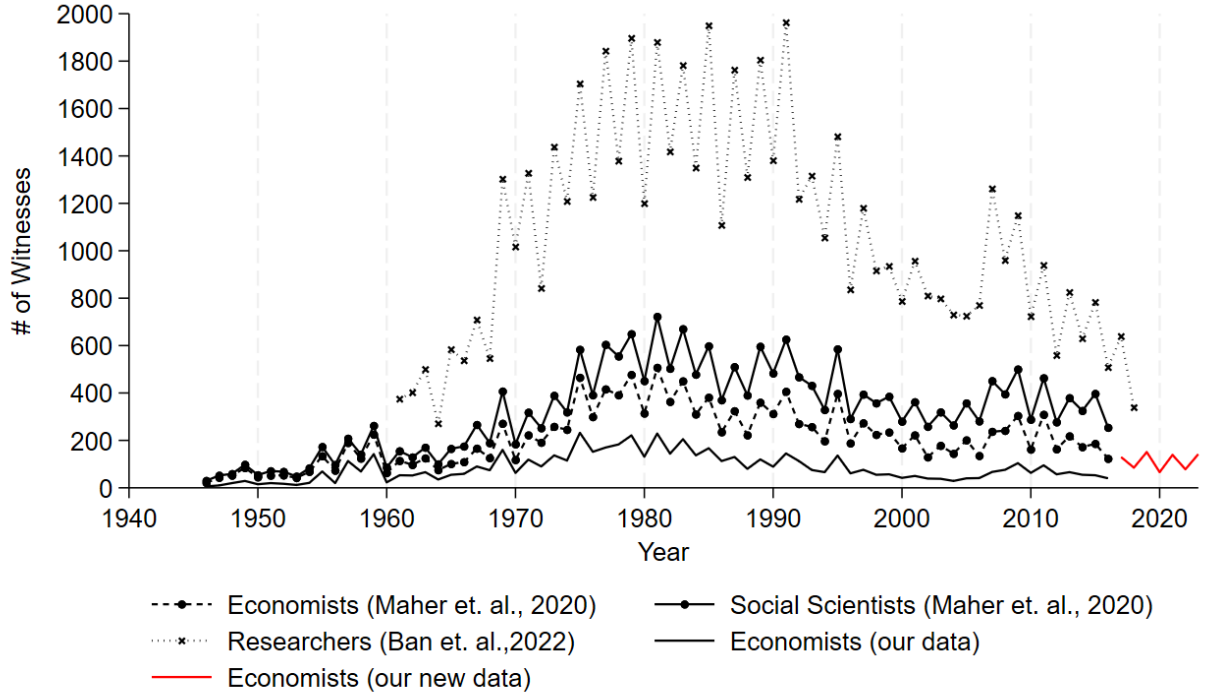


Figure 1: Number of witnesses by type and data source

Notes. Economists and social scientists, 1946-2016, are from Maher et al. (2020); economists there include those affiliated with universities, think tanks, and government agencies, not academics alone. Researchers, 1960-2018, are from Ban et al. (2023a) and include all university- and think-tank-affiliated witnesses, shown for reference only and not used in our analysis. Economists (our data), 1946-2016, restrict the Maher et al. (2020) economist series to witnesses holding a university affiliation at the time of the hearing. Economists (our new data), 2017-2023, apply the same university-affiliation restriction to witnesses identified through our GPT-4o extraction, described above.

Beyond these aggregate trends, witness activity also varies systematically by chamber. Figure 2 depicts the number of economist appearances as witnesses (panel a) and the number of hearings with witnesses (panel b), by chamber, between 1969 and 2023. The House of Representatives had more hearings with witnesses and more economist appearances for most of the years, in line with its overall larger share of hearings compared to the Senate. It is notable but expected that the number of hearings and economist appearances increased significantly during COVID-19. Congress passed numerous rescue plans to create and retain jobs, stimulus payments to individuals and businesses, expanded healthcare for certain groups, and more ⁶. The severe economic downturn, including increases in unemployment and business closures, likely increased both the need for more hearings and the use of economists' expertise to navigate complex policy development and ensure evidence-based decision making.

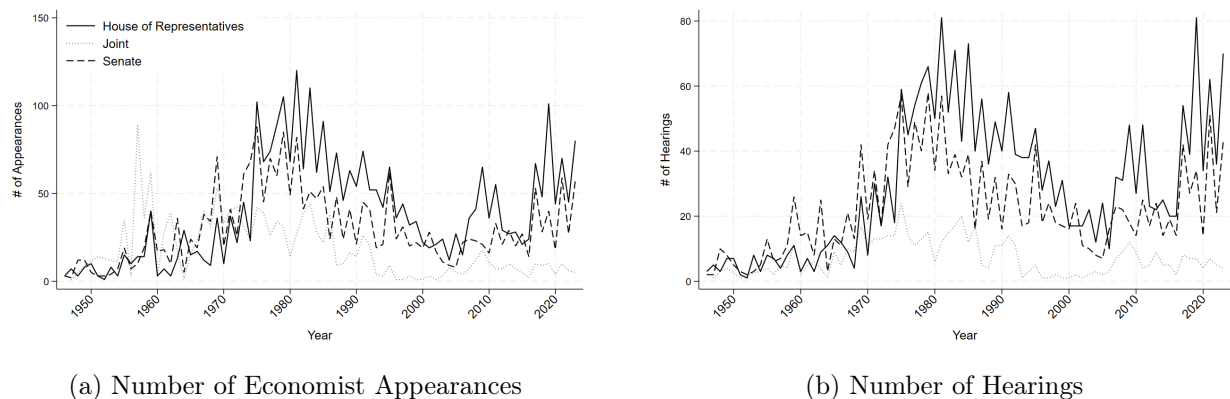


Figure 2: Number of Economist Appearances and Hearings with Economists as Witnesses, by Chamber

Notes. Economist counts reflect the number of witness appearances by economists, not unique individuals; an economist testifying in multiple hearings is counted once per appearance.

Figure 3 complements this trend by summarizing the composition of hearings in which the economists in our sample appeared, by chamber and committee jurisdiction. Forty-nine percent were held by committees of the House of Representatives, 35 percent in the Senate, and 16 percent were in joint committees, in line with previous research that found hearings in the House of Representatives to be more frequent (Ban et al., 2023a). Appearances are also concentrated in a handful of committee groups. Committees with jurisdiction over banking and finance account for the largest share, at roughly 22 percent, followed by committees covering the broader economy (19 percent) and budget and appropriations (14 percent). Notably, hearings on the economy rely disproportionately on joint committees, unlike most other committee groups, where the House and Senate typically hold hearings separately.

⁶Coronavirus Aid, Relief, and Economic Security Act (CARES Act), Pub. L. No. 116-136 (2020), <https://www.congress.gov/bill/116th-congress/house-bill/748/text>.

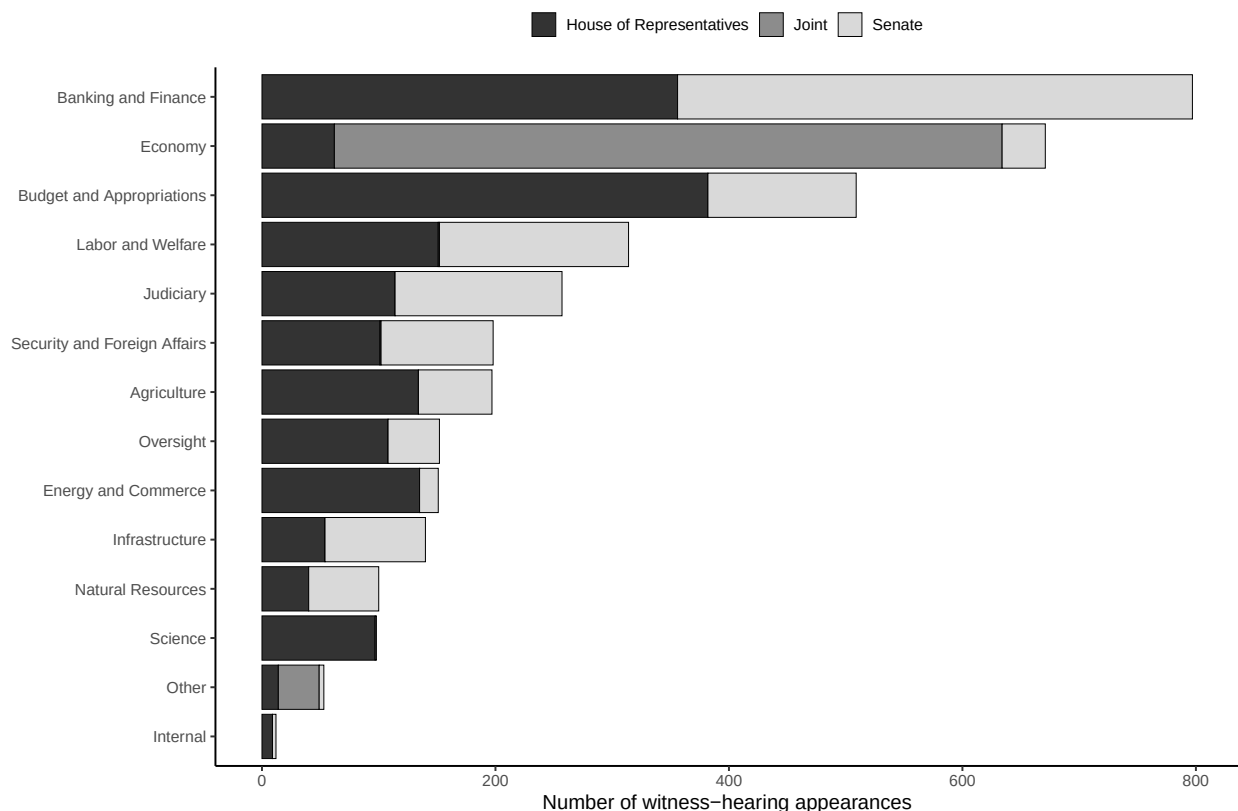


Figure 3: Distribution of Witness Appearances by Committee Group and Chamber, 1969-2023

Each bar represents one committee group, ordered by total number of witness-hearing appearances from most to least. Segments show how many of those appearances occurred in the House of Representatives, Senate, or a joint committee. The sample is restricted to witness-hearing pairs included in the final analysis panel. The same economist can appear multiple times if they testified before more than one committee. See appendix B, Table B3 for a description of our grouping of committees.

This unevenness extends beyond committees and hearing topics to individual witnesses and their institutions.⁷ Figure 4 maps witness proportions by state between 1969 and 2023, based on witnesses' university affiliation. Two facts stand out. Researchers from the East Coast are heavily represented, and there is a marked concentration around the location of top economics departments.⁸ Beyond hosting a number of elite universities, the East Coast is also better connected to Washington DC by transportation, including train service and shorter travel times from geographic proximity, making

⁷Appendix B, Table B1 lists the most frequently appearing economists between 1969 and 2023; the list includes four Nobel laureates and seven individuals who served as members, presidents, or chairmen of the President's Council of Economic Advisers, with one woman among them (Karen Davis, The Johns Hopkins School of Public Health).

⁸Appendix B, Table B2 lists the most common university affiliations, both by number of testimonies and by number of unique witnesses. By testimonies, Harvard University accounts for nearly 10 percent of testifying economists, followed by MIT (4.5 percent) and the University of Pennsylvania (3.3 percent). Notably, the University of Maryland ranks above the University of Chicago on both measures, suggesting proximity to Washington DC may matter as much as departmental prestige, while Duke University ranks comparatively low despite its strong departmental ranking and geographic proximity. Rankings by unique witnesses are similar but less extreme: MIT, for instance, accounts for 4.5 percent of testimonies but only 1.8 percent of unique witnesses.

it easier for researchers to travel to and from Congress. This pattern is consistent with a broader literature suggesting that the economics profession is characterized by a relatively homogenous elite concentrated around a handful of top departments, more so than in other fields (e.g., Ductor and Visser, 2023; Freeman et al., 2024; Korom, 2020). Aside from California, Illinois, and the upper mid-west, most US states, including large and populous ones such as Texas and Florida, have historically been underrepresented among congressional witnesses.



Figure 4: Geographic distribution of economists serving as expert witnesses in congressional hearings, by institution (1969–2024)

Note: The figure plots the geographic distribution of economists testifying as expert witnesses before U.S. congressional committees, 1969–2024. Each bubble represents a city, with size proportional to the number of testimonies given by witnesses whose institutional affiliation is located there; multiple institutions in the same city are aggregated into a single bubble on the national map. Given the concentration of witness in the Washington, DC–Boston corridor, this region is broken out separately at the institution level in the inset panel. The data from 1969–2016 is from Maher et al. (2020) and the data from 2017–2023 is from our data collection, as detailed above.

4.2 At-risk population

Congress does not systematically and publicly document who was invited to testify and declined, or who was considered and passed over, only who actually appeared. This absence poses a challenge for estimating gender gaps in testimony, since establishing whether women are underrepresented requires comparing witnesses against the pool of academics who could plausibly have been called, not simply against those who were. To address this, we introduce a novel approach to constructing this at-risk population for each witness appearance.

Common approaches to constructing at-risk populations rely on observable traits such as academic age, gender, affiliation, or publication and citation counts (e.g. Rose and Baruffaldi, 2025). We move away from this in favor of topical, or semantic, matching, identifying economists who are proximate

to the witness in the substantive content of their research. We take this approach because domain expertise plausibly sits at the heart of how witnesses are actually selected, as committees seek out testimony from those who can speak credibly to the topic at hand, not from those who merely resemble past witnesses on observable characteristics. Concretely, we construct an at-risk population for each witness appearance based on the stock of knowledge a candidate had produced by the time of the hearing.



Figure 5: Construction of the At-Risk Comparison Group

This chart summarizes the five-step pipeline used to construct the at-risk comparison group.

For each of the 4,535 witness appearances with available publication records, we identified the witness’s stock of publications in the five years prior to the hearing, averaging 13 publications per witness. We queried this publication stock through LogicMill, a tool for measuring semantic similarity between academic publications at scale, retrieving the top 500 most semantically similar publications per paper. This yielded 3,498,990 unique matched publications, with a mean cosine similarity of 0.981, confirming that retrieved papers are substantively close to the focal witness’s work⁹.

Having identified the publications most semantically proximate to each witness’s own work, we next turn from publications to people, extracting the authors of these matched publications as the candidates most proximate to the witness in expertise. This yields an initial candidate pool of 3,268,902 author records. For each witness-hearing pair, we computed each candidate’s number of appearances across matched publications as a measure of proximity intensity to the focal witness. We then applied a set of filters to retain plausible candidates: we dropped the witness themselves¹⁰, removed candidates with fewer than two appearances across matched publications, and required that candidates have at least one publication and at least one US university affiliation in the five

⁹To validate the use of the five-year publication window as a proxy for hearing-relevant expertise, we sampled all testimonies between 2017 and 2023, extracted the oral testimony given by the witness, and measured the semantic similarity between that testimony and the witness’s full publication stock, ranking publications by similarity. Appendix B, Figure B2 reports the distribution of the year distance between each witness’s most similar publication and the hearing year, separately by gender. The distribution is right-skewed and concentrated in the first decade prior to the hearing, suggesting that the most topically relevant work tends to be relatively recent, broadly consistent with the use of a five-year window. We note, however, that the distribution has a long right tail, indicating that for some witnesses the closest matching publication predates the window.

¹⁰Convincingly, witnesses were included in their associated pool of at-risk candidates in 82 percent of witness appearances.

years prior to the hearing. After filtering, the mean candidate pool per hearing stands at 607, with a minimum of 3 and a maximum of 5,252 candidates per witness. We further capped the candidate list at the top 20 ranked candidates per hearing by appearance count. This ranking favors candidates whose own publications repeatedly surface as semantically similar matches to the witness’s work, rather than candidates linked through a single matched publication alone. Of these, 47 percent met the US university affiliation criterion and were retained. This yields 21,412 unique candidate authors, for whom we retrieved full bibliographic and altmetric histories from OpenAlex and Altmetric as well as driving time between their home institution and Washington D.C. retrieved from Google Maps to construct career panels analogous to those of the actual witnesses. After constructing career panels for all at-risk individuals, we removed outliers¹¹. In our final sample, each witness-hearing appearance has a mean at-risk set of 17 individuals, median of 14, and a minimum of 5¹². After removing outliers and dropping hearings without a sufficient number of at-risk individuals, our final dataset includes 3,662 witness appearances across 2,817 hearings.

4.3 Variables

We compiled data from various data sources to create covariates at the individual and hearing level. All time-variant variables were constructed by year and included in the panel one year lagged. See Table 1 for a description of all variables.

4.3.1 Dependent Variable

Witness. Our dependent variable indicates whether an economist appeared as a witness in a given hearing, and takes the value 1 for witnesses and the value 0 for the at-risks.

4.3.2 Independent Variables

Female. The gender of each academic witness was inferred algorithmically from their first name, using the R package “gender”.¹³ In cases where gender could not be inferred (e.g., initials or non-US names) we assigned it manually by looking up the academic’s university page, LinkedIn profile, or other digital records. We were able to identify gender for 97 percent of the sample, and the remaining were dropped from our analysis. The same approach was used for the at-risk population, where gender could be inferred for 88 percent of the sample.

¹¹For example, OpenAlex falsely attributes publications to profiles in some cases, which can distort seniority, calculated as time since first publication. We drop individuals where the economist has an academic age above 65.

¹²It has been demonstrated that an at-risk to event ratio of 5:1 or higher results in little loss of efficiency in estimations (Ding et al., 2013).

¹³See: <https://cran.r-project.org/web/packages/gender/index.html>.

Virtual. We collect information about the hearing format from the processed hearing transcripts. Virtual hearings were typically conducted using Cisco Webex in the House of Representatives and the Senate, and this was announced in opening statements by committee Chairs¹⁴.

Driving Distance. To proxy mobility friction, we estimated the travel time between Washington D.C. and each academic’s home institution. Inferring this affiliation differs slightly by group. For witnesses, we draw on two sources, affiliations reported in the hearing transcript and affiliations listed on the witness’s publications. Priority is given to the affiliation reported in the hearing transcript with affiliations from the witness’s publication record used in the absence of that. In the latter case, a primary affiliation is selected whenever multiple affiliations compete in a given year using the following decision rule: First, we assign the institution appearing most frequently across the individual’s publications in that year. If a tie remains, we break it using the individual’s most common affiliation in the following year, assigning whichever tied institution matches it. If the tie persists, we prioritize U.S.-based institutions over non-U.S. institutions, after which any remaining ties are resolved by alphabetical order of institution name. For years without affiliation information from publications, we carry forward the most recent affiliation from a prior year. For at-risk candidates, no transcript affiliation exists, so location is inferred from publication affiliations alone, using the same procedure. Once an institution is assigned, we retrieve its coordinates from OpenAlex and query the Google Maps API to obtain the estimated driving time to Washington D.C.

4.3.3 Control Variables

Top 10 Department. We use the home affiliation assigned to each academic (see description above) to construct a dummy variable *Top-ten department* indicating if a given individual works for a top-ten economics department. We followed the definition of top economics departments using Amir and Knauff (2008).

Academic age. We calculate academic age using the publication record of both witnesses and at-risk candidates, drawing on OpenAlex as our data source. Academic age is constructed by subtracting the first year of publication from the year of the hearing. OpenAlex’s author disambiguation process is imperfect, and publications are occasionally attributed to the wrong author, an error that is more likely among scholars with very common names, including many Asian names. When this happens, a scholar’s profile can be mistakenly credited with a much earlier publication belonging to someone else, inflating academic age well beyond what is realistically feasible. We take two steps to address this. First, to correct for errors of this kind early in a scholar’s record, we identify all authors with a gap of more than ten years between their first and second publication, and use the second publication to assign their first-publication year instead. Second, to address remaining outliers, we drop scholars

¹⁴See example in Appendix B, Figure B1.

whose academic age exceeds 65, treating cases above this cutoff as likely disambiguation or data errors.

H-Index. Following previous studies of academics’ external engagement (Ding et al., 2013; Tartari and Salter, 2015), we proxy each academic economist’s prominence using their H-index in the year prior to the hearing, calculated from their publication and associated citations records in OpenAlex.

Top-5 publication Stock. We additionally proxy prominence using each economist’s stock of publications in top-5 economics journals, identified from their OpenAlex publication record, capturing standing within the discipline that citation-based measures alone may not fully reflect (Heckman and Moktan, 2020).

Coauthors. We proxy social connectedness by counting each economist’s unique coauthors in OpenAlex, since coauthors are considered strong ties in academia (Azoulay et al., 2010; Ding et al., 2013), and we assume a larger coauthor network increases the probability of being invited to testify.

Witness Ties. Using coauthor information from OpenAlex, we construct a measure of each economist’s ties to previous witnesses, since congressional members and staffers may draw on their existing network to identify further experts. We expect direct ties to previous witnesses to be particularly consequential for women’s probability of being selected (Ding et al., 2013).

Media Mentions. We proxy each academic’s public visibility using the number of mentions in news articles, blog posts, and podcasts associated with their stock of publications prior to the hearing. The data for this measure is drawn from [Altmetric.com](https://www.altmetric.com). This measure captures whether their work has attracted attention beyond the academic community. Because congressional staff and members may rely on highly visible research when identifying experts.

Policy Mentions. We proxy each academic’s policy relevance using the number of mentions in policy documents and clinical guidelines associated with their stock of publications prior to the hearing. The data for this measure is also drawn from [Altmetric.com](https://www.altmetric.com). This measure captures whether their work has entered policy debates directly.

Table 1: Variable Definition

Variable name	Description
<i>Witness</i>	Indicator equal to 1 if the academic appeared as a witness at the hearing, 0 if included in the matched at-risk group
<i>Female</i>	1 = female, 0 = male
<i>Virtual</i>	1 = hearing conducted in a virtual or hybrid format, 0 = in-person
<i>log(Driving Time)</i>	Driving time in hours between the academic's university affiliation and Washington DC
<i>Top 10 University</i>	1 = affiliated with a top 10 economics department, 0 = otherwise
<i>Academic age</i>	Number of years since the scholar's first publication
<i>log(Top 5 Publication Stock)</i>	Number of the academic's publications in the top-5 economics journals
<i>log(H-Index)</i>	H-index of the academic, defined as the maximum value h such that the academic has h papers each cited at least h times.
<i>log(Coauthors)</i>	Count of unique coauthors from the academic's stock of publications
<i>log(Witness Ties)</i>	Count of coauthor ties to economists who have previously testified as an expert witness
<i>log(Media Mentions)</i>	Number of media mentions associated with the academic's stock of publications
<i>log(Policy Mentions)</i>	Number of citations in policy documents and clinical guidelines associated with the academic's stock of publications

4.4 Descriptive Statistics

We now turn to descriptive statistics for the sample used in our main estimations, which pools witnesses with their corresponding at-risk economists. We observe 3,662 witness appearances from 1,326 unique witnesses in our data.¹⁵ We first summarize the gender composition of the two groups. We then compare their human capital, network, and geographic characteristics, separately

¹⁵43.7 percent of witnesses appear more than once (33.3 percent of women and 45.2 percent of men). Among repeat witnesses, the plurality testify exactly twice (39.6 percent), though 141 witnesses appear more than five times, with one witness testifying 58 times. See Appendix B, Figure B3.

by gender, to document the baseline differences that motivate our empirical analysis. We conclude with pairwise correlations among the outcome and control variables used in our main and robustness specifications.

Table 2 reports the gender composition of the at-risk and witness populations. Women account for 10.5 percent the at-risk group and 10.1 percent of witnesses, a comparable number. This indicates that the underrepresentation of women among witnesses already characterizes the broader pool of economists whose expertise would have made them plausible candidates to testify.

Table 2: Number of witnesses and at-risk economists by gender

At-Risks		Witnesses	
Female	Male	Female	Male
5101 (10.5%)	43634 (89.5%)	369 (10.1%)	3293 (89.9%)

Note: Witnesses and at-risk economists are counted once per hearing-appearance, so the same individual may be counted more than once in this table if they appear across multiple hearings.

Table 3 reports summary statistics for the academic, professional, and geographic, and geographic variables used in our analysis, separately by gender, for the at-risk and witness groups. All variables, other than *Academic Age*, *Top 10 University* and *Driving Time* are measured in the year prior to the hearing. Average H-index is similar across genders within each group, but noticeably higher among at-risk economists than among witnesses. Men hold substantially larger stocks of top-five publications, larger coauthor networks, and more ties to previously testifying witnesses, gaps that are each somewhat narrower among witnesses than among the at-risk pool. Men also have a higher academic age on average, by about three years among at-risk economists and four years among witnesses, and are more likely to be affiliated with a top-ten economics department, a gap that widens from four percentage points in the at-risk group (19 versus 23 percent) to thirteen percentage points among witnesses (17 versus 30 percent). Geographically, women are on average closer to Washington DC than men in both groups, and this gap is roughly twice as large among witnesses as among the at-risk pool. Women, by contrast, have a higher stock of media attention than men in both groups, with the gap more pronounced among witnesses, while the gap in policy attention narrows considerably among witnesses relative to the at-risk pool.

Table 4 reports pairwise Pearson correlations between the outcome variable and the control variables used in our main and robustness specifications. Consistent with the equal raw share of women reported in Table 2, the correlation between witness status and gender is close to zero, underscoring the need for the multivariate analysis that follows to detect any conditional gender gap. Being a witness is instead positively correlated with ties to previous witnesses, top-five publication stock, and top-ten university affiliation, and negatively correlated with coauthor count and driving time to Washington DC, though most of these correlations are modest in magnitude. Female is negatively

Table 3: Summary Statistics, for at-risk and witness appearances

Variable	Group	Gender	N	Mean	SD	Median	Min	Max
H-Index	At-Risk	Female	5101	13.55	14.67	9.00	0.00	117.00
		Male	43634	13.68	15.16	9.00	0.00	151.00
	Witness	Female	369	11.91	11.25	8.00	0.00	64.00
		Male	3293	12.45	14.27	8.00	0.00	154.00
Top 5 Publication Stock	At-Risk	Female	5101	1.47	3.12	0.00	0.00	31.00
		Male	43634	4.87	9.88	0.00	0.00	84.00
	Witness	Female	369	1.53	2.71	0.00	0.00	20.00
		Male	3293	5.53	10.11	2.00	0.00	84.00
Coauthors	At-Risk	Female	5101	74.23	99.67	42.00	0.00	1075.00
		Male	43634	90.92	126.04	49.00	0.00	1142.00
	Witness	Female	369	59.83	91.10	29.00	0.00	1141.00
		Male	3293	72.20	103.05	40.00	0.00	1093.00
Witness Ties	At-Risk	Female	5101	1.63	3.44	0.00	0.00	35.00
		Male	43634	2.49	5.81	0.00	0.00	76.00
	Witness	Female	369	3.12	5.13	1.00	0.00	34.00
		Male	3293	3.62	7.31	1.00	0.00	76.00
Academic Age	At-Risk	Female	5101	22.71	11.67	21.00	1.00	65.00
		Male	43634	25.90	13.47	24.00	1.00	65.00
	Witness	Female	369	21.33	11.01	20.00	2.00	64.00
		Male	3293	25.61	13.26	24.00	1.00	65.00
Top 10 University	At-Risk	Female	5101	0.19	0.39	0.00	0.00	1.00
		Male	43634	0.23	0.42	0.00	0.00	1.00
	Witness	Female	369	0.17	0.38	0.00	0.00	1.00
		Male	3293	0.30	0.46	0.00	0.00	1.00
Driving Time (hours)	At-Risk	Female	5101	12.00	12.20	7.37	0.03	42.55
		Male	43634	13.74	12.99	7.85	0.01	69.08
	Witness	Female	369	8.34	10.89	5.78	0.04	44.00
		Male	3293	11.16	11.56	7.25	0.03	66.77
Policy Mentions	At-Risk	Female	5101	33.36	114.17	1.00	0.00	1843.00
		Male	43634	24.20	128.25	0.00	0.00	3780.00
	Witness	Female	369	27.75	75.41	1.00	0.00	757.00
		Male	3293	25.53	127.94	0.00	0.00	3126.00
Media Mentions	At-Risk	Female	5101	19.11	97.90	0.00	0.00	1705.00
		Male	43634	7.96	82.09	0.00	0.00	4153.00
	Witness	Female	369	28.57	134.31	0.00	0.00	1774.00
		Male	3293	9.05	85.35	0.00	0.00	2107.00

Summary statistics are captured for each individual economist-hearing pair at the time of the hearing. The same economist can appear as an at-risk or witness across multiple hearings, and in that case will be included multiple times in the summary statistics.

correlated with coauthor count, witness ties, academic age, top-five publication stock, and driving time, and positively correlated with policy attention, media attention, and whether the hearing was held virtually, mirroring the gender patterns reported in Table 3.

Table 4: Correlation Matrix of Outcome and Control Variables

Variable	1	2	3	4	5	6	7	8	9	10	11	12
1. Witness	1.00											
2. Female	-.00	1.00										
3. log(H-Index)	.01*	-.00	1.00									
4. log(Coauthors)	-.05***	-.05***	.61***	1.00								
5. log(Witness Ties)	.08***	-.04***	.54***	.41***	1.00							
6. Academic age	-.01*	-.07***	.39***	.60***	.26***	1.00						
7. log(Top 5 Publication Stock)	.05***	-.12***	.39***	.17***	.48***	.21***	1.00					
8. Top-10 University	.04***	-.03***	.18***	.11***	.21***	.06***	.31***	1.00				
9. log(Driving Time)	-.07***	-.06***	-.07***	-.02***	-.07***	-.04***	-.08***	.07***	1.00			
10. log(Policy Mentions)	-.00	.10***	.62***	.43***	.50***	.31***	.13***	.08***	-.05***	1.00		
11. log(Media Mentions)	.01***	.12***	.36***	.29***	.23***	.18***	.01*	.03***	-.03***	.61***	1.00	
12. Virtual	.01	.09***	.15***	.12***	.06***	.09***	-.04***	-.03***	-.02***	.28***	.50***	1.00

Pearson correlations, computed pairwise on non-missing observations. Numbered columns correspond to the numbered variables listed in the first column. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

5 Methods

A central challenge in studying the role of mobility constraints in academic engagement with policy advising is endogeneity. Economists who testify are potentially selected on unobservable characteristics correlated with both gender and perceived mobility costs. For example, women who reach the witness pool despite structural barriers may be more prominent or more mobile than the average female economist, which would bias naive comparisons toward understating the true mobility constraint. Relatedly, an economist’s institutional location is itself a choice, potentially correlated with unobserved factors that also shape their propensity to engage in advisory work, so that any association between distance and testimony propensity may partly reflect sorting into location rather than a causal effect of distance itself. We address these concerns by exploiting the shift to virtual hearings induced by the COVID-19 pandemic as a source of variation in mobility costs that was driven by the public health crisis, as described in section 3.2, where the requirement to appear in-person was eliminated, and replaced by a virtual format. This change, we propose in hypothesis 5, reduced effective travel costs disproportionately for women located farther from Washington.

Our main analysis estimates a conditional logit model to examine the probability that an economist is selected to testify before Congress relative to a matched pool of plausible alternatives. For every hearing-witness, we construct a stratum consisting of the realized witness and her matched at-risk economists identified through the semantic similarity procedure described in section 4.2. The unit of analysis is therefore the academic-hearing pair, where the dependent variable, Y_{ih} , equals one if academic i appeared as a witness for hearing h , and zero otherwise.

Formally, the conditional probability that economist i is selected as a witness in hearing h is given by:

$$P(Y_{ih} = 1 | h) = \frac{\exp(X_{ih}\beta + \eta_h)}{1 + \exp(X_{ih}\beta + \eta_h)} \quad (1)$$

where X_{ih} denotes economist-level characteristics at the time of the hearing and η_h denotes the hearing strata. Because the model conditions on the hearing stratum, all hearing-level characteristics are absorbed by the conditional likelihood. Identification therefore comes entirely from within-hearing variation between the selected witness and the matched at-risk economists associated with the same hearing. This design controls for both observed and unobserved hearing-specific factors, including committee composition, political salience, hearing topic, and the overall demand for expert testimony, including any demand shifts toward subfields where women are overrepresented.

The fully interacted estimating equation is specified as:

$$\begin{aligned} X_{ih}\beta = & \beta_1 Female_i + \beta_2 Virtual_h + \beta_3 \log(DrivingTime_{ih} + 1) + \beta_4 Female_i \times Virtual_h \\ & + \beta_5 Female_i \times \log(DrivingTime_{ih} + 1) + \beta_6 Virtual_h \times \log(DrivingTime_{ih} + 1) \quad (2) \\ & + \beta_7 Female_i \times Virtual_h \times \log(DrivingTime_{ih} + 1) + \gamma Z_{ih} \end{aligned}$$

where $Female_i$ is an indicator equal to one for women economists, $Virtual_h$ indicates hearings conducted during the COVID-19 period in which hearings were held virtually, and $\log(DrivingTime_{ih})$ measures the logged driving time from the economist's institutional affiliation to Washington DC. The vector Z_{ih} includes controls for H-index, cumulative number of coauthors, ties to previous witnesses, academic age, top-ten university affiliation, stock of top-five publications, and stocks of media and policy attention. All controls, other than top-ten university affiliation and academic age, are log transformed and measured in the year prior to the hearing.

The coefficient of the interaction between female and driving time (β_5) captures whether geographic distance imposes a larger penalty on women economists relative to men in the selection process, as proposed in hypothesis 3. Our interpretation of the triple interaction coefficient (β_7) rests on the assumption that virtual hearings lower the cost of participation disproportionately for geographically distant candidates, making this highly visible, high-impact dissemination opportunity more attainable to distant women economists who might otherwise face binding mobility constraints. The triple interaction examines whether the transition to virtual hearings reduced this differential distance penalty for women, as proposed in hypothesis 5. A positive β_7 coefficient would indicate that virtual hearings disproportionately increased the likelihood that geographically distant women economists were selected to testify, consistent with the interpretation that mobility constraints constitute an important barrier to women's participation in congressional testimony.

Because the hearing fixed effects η_h are conditioned out of the likelihood rather than estimated directly, predicted probabilities and average marginal effects are not identified in this framework. We therefore report and interpret coefficients in log-odds terms throughout. Standard errors are

clustered at the hearing level, since candidates from the same hearing share a common choice set and *Virtual* is constant within each hearing.

6 Results

We test our hypotheses using the conditional logit framework described in Section 5. Specifically, we examine whether female economists are less likely than comparable male economists to be selected as expert witnesses (Hypothesis 1), whether geographic distance from Washington DC is negatively associated with the odds of appearing as a witness (Hypothesis 2), whether this distance penalty is stronger for women (Hypothesis 3), whether the shift to virtual hearings increased women’s odds of selection more than men’s (Hypothesis 4), and whether this format effect was larger for women located farther from Washington DC (Hypothesis 5).

Table 5 presents estimates from a series of conditional logit models in which each hearing defines a matched stratum consisting of the economist selected to testify and the pool of at-risk economists associated with that appearance. The dependent variable indicates whether an economist appeared as a witness, and identification relies entirely on within-stratum variation; because comparisons are made only among economists eligible for the same hearing, all hearing-level characteristics are absorbed by construction. Coefficients throughout are reported as log-odds.

Table 5: Effect of Geographic Proximity, Hearing Format, and Network on Odds of Testifying

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female	-0.06 (0.06)	-0.06 (0.06)	-0.11* (0.06)	0.54*** (0.15)	-0.18*** (0.06)	0.58*** (0.16)	0.84*** (0.21)
log(Driving Time)			-0.31*** (0.02)	-0.27*** (0.02)	-0.31*** (0.02)	-0.27*** (0.02)	-0.28*** (0.03)
Female x log(Driving Time)				-0.34*** (0.08)		-0.40*** (0.08)	-0.62*** (0.11)
Female x Virtual					0.84*** (0.23)	-0.09 (0.54)	
Virtual x log(Driving Time)						-0.16 (0.15)	
Female x Virtual x log(Driving Time)						0.49** (0.23)	
Female x log(Witness Ties)							-0.39** (0.17)
log(Witness Ties) x log(Driving Time)							0.00 (0.02)
Female x log(Witness Ties) x log(Driving Time)							0.36*** (0.08)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	52397	52397	52397	52397	52397	52397	52397
Testimonies	3662	3662	3662	3662	3662	3662	3662
Clusters	2817	2817	2817	2817	2817	2817	2817
Log-likelihood	-9651.25	-8873.05	-8766.80	-8754.07	-8759.76	-8744.33	-8732.24

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

All models are estimated by conditional logit within a matched set, where each matched set consists of the witness and the at-risk economists associated with a given hearing. Coefficients are reported as log-odds. Standard errors in parentheses are clustered at the hearing level. The dependent variable is an indicator for whether the economist appeared as an expert witness in a given hearing. Estimation by conditional logit models. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$. Full table with control coefficients can be found in Appendix B, Table B4.

Models 1 through 3 test Hypothesis 1: Model 1 excludes controls, Model 2 adds the full set of controls, and Model 3 further adds driving distance to Washington. In the first two specifications, the coefficient on Female is small and not statistically significant, indicating that, conditional on being in the choice set for a given hearing, women are no less likely than men to appear as a witness, whether or not we account for differences in academic and professional characteristics between women and men. Once driving time to Washington D.C. is added as a control in Model 3, the coefficient on Female turns negative and statistically significant. Women have approximately 10 percent lower odds of appearing as a witness than comparable men once distance is accounted for. Taken together, we find only partial support for Hypothesis 1.

Model 3 retains these controls and adds the logged driving time between the economist's institution and Washington DC, testing Hypothesis 2. Driving time enters negatively and significantly, indicating that being farther from Washington lowers the odds of appearing as a witness for all academics, women and men alike, while the coefficient on Female remains statistically insignificant.

We therefore accept Hypothesis 2, finding that geographic distance reduces the odds of appearing as a witness for all economists, though not yet differentially by gender.

Model 4 adds an interaction between *Female* and log driving time to test Hypothesis 3, whether women’s odds of testifying decline more steeply with distance than men’s. The interaction coefficient is negative and highly significant, so distance lowers the odds of testifying for all economists but significantly more so for women, supporting the hypothesis. The main effect of *Female* in this model, which now captures the gender gap among economists located closest to Washington DC, is positive and significant. For economists closest to the capital, a woman’s odds of appearing as a witness are approximately 1.7 times¹⁶ those of a comparable man drawn from the same hearing’s choice set. This advantage narrows by 0.34 log-points for each additional unit of log driving time, so that women’s overall distance slope is -0.61 ¹⁷, more than twice as steep as the corresponding slope for men (-0.27). The right panel of Figure 6 plots this pattern directly, showing a modest female premium within four hours of Washington that turns into a growing female penalty from roughly seven hours onward.

Model 5 tests Hypothesis 4 with a different specification, dropping the distance interaction and instead interacting *Female* with hearing format. The main effect of *Female* here, capturing the gender gap under in-person hearings, is not statistically significant, indicating no in-person gender gap in selection. The interaction between *Female* and *Virtual* is large, positive, and highly significant, increasing the female-to-male odds ratio by a factor of roughly 2.3 relative to the in-person baseline.¹⁸ This supports the hypothesis. The left panel of Figure 6 shows this shift directly, with the female gap statistically indistinguishable from zero in person and positive and significant once hearings move online.

¹⁶The reported odds ratio is obtained by exponentiating the *Female* coefficient from Model 4. Because the specification includes an interaction between *Female* and log driving time, the coefficient on *Female* alone captures the gender gap evaluated at $\log(\text{DrivingTime} + 1) = 0$, that is, among economists located closest to Washington DC. Exponentiating this coefficient converts the log-odds difference into an odds ratio: $e^{0.54} \approx 1.72$.

¹⁷ $-0.27 + (-0.34) = -0.61$.

¹⁸The main effect of *Virtual* itself drops out of the model, as this hearing-level characteristic is constant within a stratum and is therefore absorbed by the hearing fixed effect.

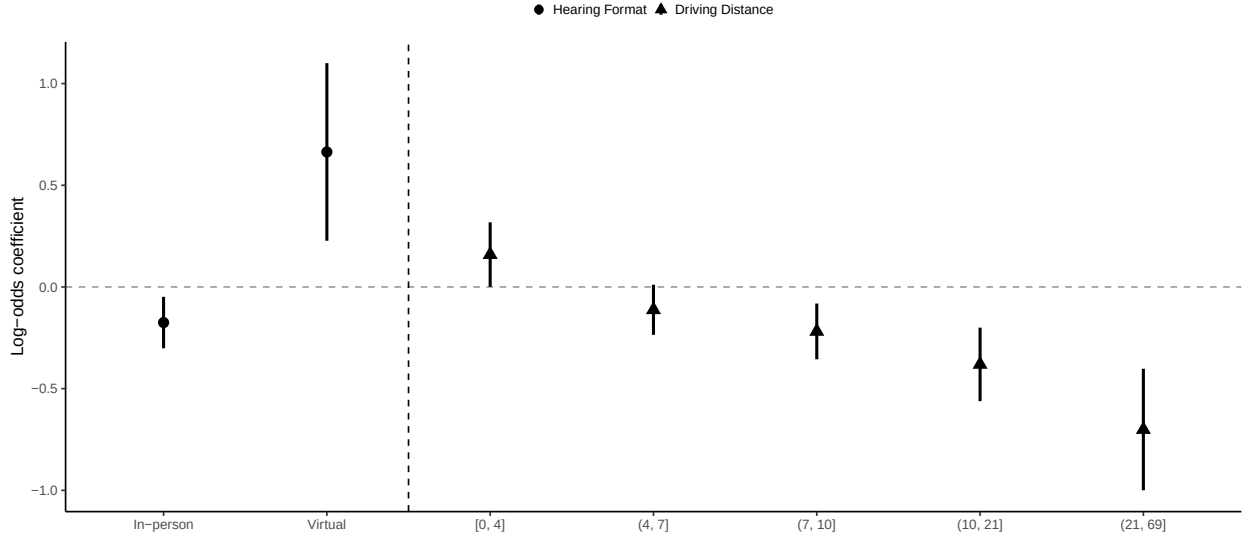


Figure 6: Gender Gap by Hearing Format and Distance to Washington D.C.

Left panel: The gender gap in the log-odds of appearing as an expert witness before Congress, by in-person and virtual hearing format. The plotted coefficients are the interaction between $Female \times Virtual$, corresponding to Model 5 in Table 5. Right panel: The gender gap in log-odds of testifying by driving distance between the economist's affiliation and Washington DC, from an interaction between $Female \times \log(Driving\ Time)$, corresponding to Model 4 in Table 5. Coefficients are evaluated at the mean driving time within each of five quantile-based ranges of driving time to Washington DC. Square brackets indicate the boundary value is included in the range, parentheses indicate it is excluded. Range boundaries are rounded to the nearest whole hour for readability. Because the model is linear in log driving time, this is equivalent to the average predicted effect across all economists in that range, rather than a single representative point. The vertical bars represent 95 percent confidence intervals.

Model 6 combines both interactions in a single fully interacted specification, adding the triple interaction between Female, driving time, and hearing format to test Hypothesis 5. The logic is that if geographic mobility friction is the constraint actually binding women's participation in in-person hearings, the shift to virtual format should produce the largest gains for women located farthest from Washington. The triple interaction coefficient is positive and significant, supporting the hypothesis. One coefficient changes in an instructive way once distance and format are interacted jointly. The double interaction between Female and *Virtual*, significant on its own in Model 5, becomes small and statistically insignificant in Model 6. This does not contradict Hypothesis 4; rather, it shows that the average format effect estimated in Model 5 is not uniform across distances. Once the triple interaction lets this effect vary by distance, the format advantage turns out to be concentrated among women located farther from Washington, and close to zero for women who were already near the capital, precisely the pattern Hypothesis 5 predicts. By contrast, the double interaction between Female and log driving time remains negative, significant, and similar in magnitude to Model 4, indicating that the baseline distance penalty for women is not simply an artifact of pooling across hearing formats.

Model 7 does not test an additional hypothesis. Model 7 serves instead as a reference specification, replacing the virtual-format interactions with an analogous set of interactions using ties to previously testifying witnesses. Professional networks are a well-documented channel through which career opportunities are allocated (Granovetter, 1973; Dustmann et al., 2016), making prior witness ties a natural benchmark against which to evaluate the virtual format effect. As reported in Table 5, the triple interaction between Female, log driving time, and prior witness ties is positive and significant, similar in sign to the corresponding triple interaction in Model 6. On its own, this could suggest that relevant network ties moderate the distance penalty in much the same way that hearing format does. Figure 7 tests this impression more directly by plotting the predicted Female coefficient implied by Models 6 and 7 together, at short and long driving distances.

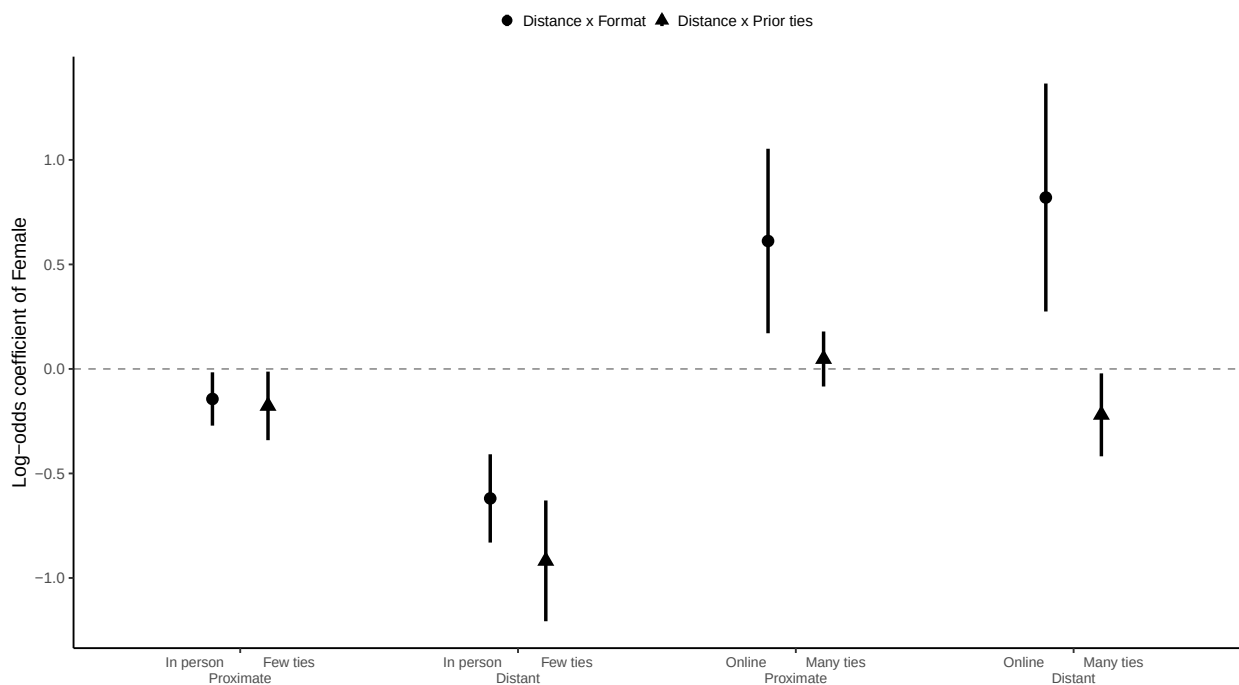


Figure 7: Gender Gap by Distance to Washington DC, Hearing Format, and Prior Witness Coauthor Ties

The gender gap in likelihood to testify by driving distance, hearing format, and prior witness ties. Coefficients for hearing format are derived from the triple interaction between *Female* $\times$ $\log(\text{Driving Time})$ $\times$ *Virtual* (Model 6 in Table 5); coefficients for prior witness ties are derived from the corresponding triple interaction between *Female* $\times$ $\log(\text{Driving Time})$ $\times$ *Witness ties* (Model 7 in Table 5). Estimates are evaluated at the 25th (proximate) and 75th (distant) percentile driving distance from Washington DC, separately by hearing format and by level of prior witness ties.

In each model's own reference category, in-person hearings in Model 6 and few prior ties in Model 7, the female coefficient follows a similar pattern despite coming from two separate estimations: close to null for economists near Washington and significantly negative for those farther away. Moving to the corresponding treatment condition, the two mechanisms tell a different story. A large stock of prior

ties has no distinguishable effect on the female coefficient for economists near Washington, and has a negative effect for economists far from Washington D.C. A virtual format, by contrast, produces a substantially larger increase in the female coefficient for economists far from Washington, one that clearly outpaces what a strong network of ties delivers at the same distance, and this comparison does not even weigh the relative cost of the two interventions, prior coauthorship ties accumulate slowly over a career and are not easily created by policy, whereas moving a hearing online is a single organization design choice available immediately.

6.1 Robustness checks

We conduct a series of robustness checks to evaluate whether our main results depend on specific modeling or sample construction choices. We first implement a placebo test that directly probes the causal logic behind our virtual-hearing results, then examine sensitivity to the time period covered, the size of the at-risk pool, and the treatment of outliers in our main sample. Across these checks, the baseline distance penalty for women and its interaction with prior witness ties prove highly robust, while the direct interaction between gender and virtual format is more sensitive to sample composition.

Our identification strategy relies on the assumption that the shift to virtual hearings during the COVID-19 pandemic provides a source of variation in hearing format, unrelated to the underlying selection process. To assess whether our estimated interaction effects could instead arise mechanically, for instance from unobserved differences between hearings that happened to occur online, or due to time trends, we implement a placebo test that simulates a counterfactual shift to virtual format among hearings held before the pandemic, when no hearings were conducted virtually.

We proceed in three steps. First, we calculate the actual proportion of hearings conducted virtually during the pandemic, separately by chamber. Second, we use these proportions to randomly assign a placebo *Virtual* indicator to hearings held before 2020, so that the chance of each pre-pandemic hearing to be treated as virtual has a probability matching its chamber’s actual virtual-hearing rate during the pandemic. Third, we re-estimate Model 5 and Model 6 on this placebo sample, recording the coefficient on $Female \times Virtual$, testing Hypothesis 4, and the triple interaction $Female \times \log(Driving\ Time) \times Virtual$, testing Hypothesis 5. We repeat this procedure 500 times, drawing a new independent placebo assignment of *Virtual* on each iteration, and plot the resulting distribution of each coefficient across the 500 draws.

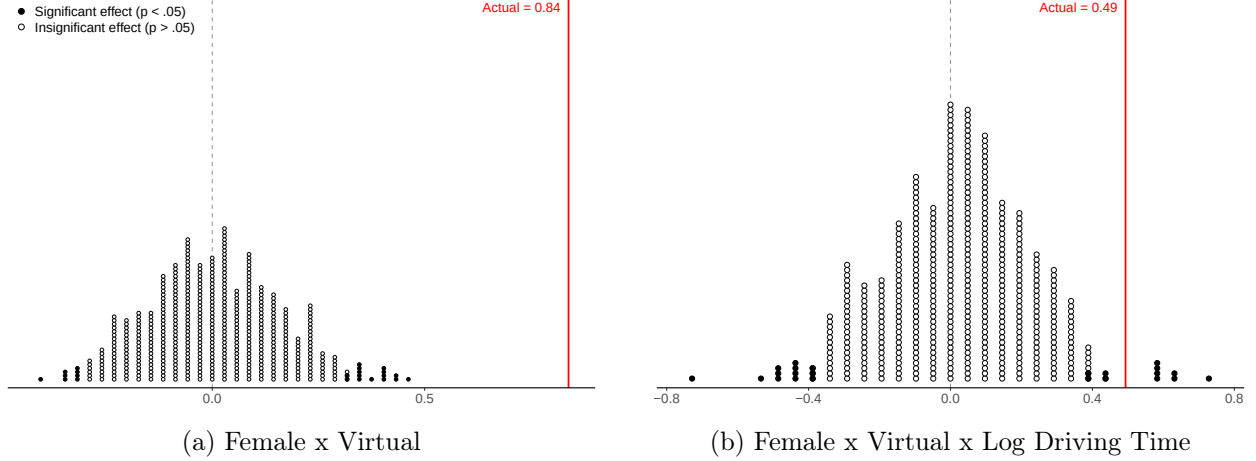


Figure 8: Placebo Distributions of the $Female \times Virtual$ and $Female \times Virtual \times \log(Driving\ Time)$ Coefficients

Notes. Distribution of the coefficient of interest from 500 placebo estimations, in which a placebo *Virtual* indicator is randomly assigned to pre-pandemic hearings using chamber-specific virtual-hearing rates observed during the pandemic. To avoid contamination, the pandemic years 2020-2023 where the actual treatment occurred are excluded from the placebo tests. Panel (a) shows the $Female \times Virtual$ coefficient from Table 5 Model 5 (23 of 500 draws individually significant at the 5 percent level). Panel (b) shows the $Female \times Virtual \times \log(Driving\ Time)$ coefficient from Table 5 Model 6 (23 of 500 draws individually significant at the 5 percent level). The dashed vertical line in each panel marks the corresponding coefficient estimated on the actual sample in Table 5.

Figure 8 presents the results. Panel (a) shows that the placebo distribution of the $Female \times Virtual$ coefficient is centered near zero, while our main estimate falls far in its right tail. None of the 500 placebo draws produce a coefficient as large as our actual estimate, indicating that an effect this large is unlikely to arise by chance alone. Panel (b) shows the same pattern for the triple interaction coefficient. Its placebo distribution is likewise centered near zero, and our main estimate again falls in the tail, exceeded by only seven of the 500 draws. In both panels, the share of individual placebo draws that would themselves register as significant at the 5 percent level sits close to that nominal rate (4.6 percent), confirming that the placebo procedure behaves as a well-calibrated null against which to judge our main estimates. Both the direct interaction between gender and virtual format, testing Hypothesis 4, and its distance-moderated extension, testing Hypothesis 5, are therefore unlikely to have arisen by chance.

Beyond this placebo test, we assess sensitivity to three choices made in constructing the estimation sample: the time period covered, the size of the matched at-risk pool, and the treatment of outliers. Table 6 summarizes these checks, grouped by test type.

Table 6: Summary of Robustness Checks

Specification		Obs.	H1	H2	H3	H4	H5	Reference
Main specification (full sample)		52,397	✓	✓	✓	✓	✓	Table 5
Placebo test		50,131	–	–	–	✓	✓	Figure 8
Period covered	Hearings from 2000 onward	14,301	✗	✓	✓	✓	✓	Table. C1
	Hearings from 2010 onward	8,380	✗	✓	✗	✓	✗	Table. C2
At-risk pool size	Top 5 candidates	21,972	✓	✓	✓	✓	✓	Table. C3
	Top 10 candidates	39,315	✓	✓	✓	✓	✓	Table. C4
	Top 15 candidates	50,643	✓	✓	✓	✓	✓	Table. C5
Including outliers		76,904	✓	✓	✓	✓	✓	Table. C6

✓ indicates the hypothesis is supported in this specification; ✗ indicates it is not. Hypothesis 1 is evaluated from the *Female* coefficient in Model 1; Hypothesis 2 from $\log(\text{Driving Time})$ in Model 3; Hypothesis 3 from $\text{Female} \times \log(\text{Driving Time})$ in Model 4; Hypothesis 4 from $\text{Female} \times \text{Virtual}$ in Model 5; Hypothesis 5 from $\text{Female} \times \text{Virtual} \times \log(\text{Driving Time})$ in Model 6. All models are estimated by conditional logit within a matched set, where each matched set consists of the witness and the at-risk economists associated with a given hearing.

Across these checks, the results are broadly robust, with support for Hypothesis 1 being the weakest. In particular, the gender gap conditional on individual-level characteristics is not supported in the models that restricts the year range to more recent years. This could be explained by some of the supply side factors, such as the alleviation of some structural barriers to women’s participation, or demand side factors, such as increased effort to promote diversity in Congressional operations¹⁹. Hypotheses 2 and 4 hold in every specification without exception. Hypotheses 3 and 5 hold in five of the six checks, failing only in the 2010-onward restriction, the specification with by far the fewest observations (8,096 against 52,397 in the main specification). Notably, both hypotheses fail together in the 2010-onward sample. This pattern is consistent with reduced statistical power in the smallest sub-samples, rather than a substantive breakdown of the underlying relationships.

¹⁹One example is the Congressional Tri-Caucus initiative, announced in 2019 and launched in 2020, which invited House committees to survey the demographic diversity of outside witnesses. The initiative was voluntary and House-specific, imposed no selection requirements or demographic targets, and does not appear to have been institutionalized as a continuing chamber rule or program. We are not aware of any mandatory House-, Senate-, or Congress-wide program governing the demographic composition of committee witnesses (Congressional Tri-Caucus, 2019).

7 Discussion and Conclusion

Our paper asks whether geographic mobility friction helps explain gender disparities in experts’ participation in advisory work with policymakers, and provides causal evidence on the effect of removing this friction. Expert advisory work, from scientific advisory boards and consulting to public testimony, tends to concentrate geographically in a small number of hubs, such as corporate headquarters or the seat of government, and typically requires physical presence. The burden of traveling to reach them is known to fall unevenly across individuals, weighing more heavily on women.

We study this question using academic economists’ testimony before the U.S. Congress. This setting combines a consequential and highly visible form of expert engagement with an unusually well documented public record and an abrupt, exogenous change in mobility requirements. We construct a dataset capturing the universe of economists who testified between 1969 and 2023, matched to a hearing-specific pool of at-risk candidates. Conditional on individual-level characteristics and driving time to Washington, women have around 10 percent lower odds of appearing as an expert witness, though this gap is sensitive to the sample period. It disappears in hearings held after 2000, consistent with Congress’s own efforts to broaden diversity among invited experts. Among economists closest to the capital, women are in fact *more* likely than comparable men to appear as a witness, but a disadvantage for remote women emerges which widens with driving distance, the pattern that drives the average gap reported above. The COVID-19 pandemic forced hearings from in-person to virtual formats, effectively removing the requirement for physical presence. Women saw disproportionate gains to their odds of appearing as an expert witness, and these gains were concentrated among those located farthest from Washington, the group for whom the presence requirement had been most binding. Together these findings identify geographic mobility friction as a distinct and previously overlooked supply-side mechanism underlying gender gaps in access to expert advisory work.

The comparison between our virtual-hearing result and two related studies helps locate what the mobility friction actually is. Ding et al. (2010) finds that lowering communication costs through information technology benefited women scientists’ productivity and collaboration more than men’s. Catalini et al. (2020) shows that cheaper air travel, which lowered the cost of a trip without removing the need to take it, benefited only male scientist pairs, with no corresponding gain for pairs involving a woman. Our design holds the advisory task itself fixed and removes only the requirement to be physically present. In doing so, it isolates that requirement, not the cost of meeting it, as the binding constraint. This distinction matters because the underlying asymmetry it exploits is not resolved by making travel cheaper. Women’s disproportionate caregiving responsibilities, and their correspondingly greater sensitivity to rigid scheduling, are well documented (Goldin, 2014; Kleven et al., 2019; Gagliardi et al., 2024). A subsidized flight to Washington still asks a witness to be

away from home on a fixed date. Removing the need to travel at all relieves exactly the constraint that caregiving asymmetries impose.

The same logic applies to how a formal, scheduled advisory relationship is understood. Audretsch and Stephan (1996) distinguish scientists who is the founder of a firm, from those who serve on its scientific advisory board, arguing that proximity matters less once the relationship is scheduled and planned rather than incidental. Their argument is that a board member does not need to live near the firm, since the interaction does not depend on daily contact the way founding a firm does. But being scheduled in advance is not the same as being remote. A board member still has to travel to the meeting on the day it is held, and it is this requirement, not the lack of daily proximity, that continues to impose a cost on those invited. That could explain why the gains we document appear once presence, and not cost, is taken off the table.

We contribute to research on gender and academic engagement, which has documented persistent gaps in academics' commercial activity. This includes patenting (Whittington and Smith-Doerr, 2008), university-industry collaboration (Tartari and Salter, 2015), academic entrepreneurship (Abreu and Grinevich, 2017), and appointment to science advisory boards (Ding et al., 2013). It was not obvious *ex ante* that a similar gap would appear in women's engagement in expert advising. The crucial role of social networks in generating commercial opportunities is well documented (Murray and Graham, 2007), particularly for opportunities that arise by invitation (Ding et al., 2013), a pattern that would predict a similar disadvantage for women in testifying. Yet several features of the public-policy setting cut the other way. Women are overrepresented in the public sector relative to the industry sectors relevant to commercialization, which could work in their favor if that translates into more invitations to testify ²⁰. Further, academics' engagement with the third sector shows the opposite pattern from commercialization, with women more likely than men to interact with charities and NGOs (Lawson and Salter, 2023). Women are also more likely to take on low-promotable assignments such as committee service, the kind of unrewarded, invitation-based work that testimony might resemble (Babcock et al., 2017). Taken together, the literature on academic engagement predicts a gender gap, but it does not settle whether that gap should extend to this particular form of engagement.

Against this backdrop, our findings provide some evidence that women have lower odds of appearing as expert witnesses, indicating that the gap in industry-facing engagement extends to the channels through which expertise reaches federal policymakers. But the gap is modest, shows indications of converging over time, and even reverses for the economists geographically closest to Washington. This convergence distinguishes policy advising from the more persistent gaps documented in commercialization, and is consistent with features of the public-sector setting that may make it comparatively more open or attractive to women.

²⁰OECD, *Government at a Glance 2025* (OECD Publishing, 2025), <https://doi.org/10.1787/0efd0bcd-en>.

Beyond academic engagement, our findings also speak to the literature on geographic proximity and knowledge transfer. Extant work documents the importance of proximity for collaboration and knowledge transfer (Jaffe et al., 1992; Katz, 1994), with recent evidence isolating the physical presence of a knowledge source itself as a causal driver of localized spillovers (Balsmeier et al., 2026). Women tend to face more mobility friction than men. This is evident in their sorting into jobs that offer better work-life balance (Barbulescu and Bidwell, 2013), greater temporal flexibility (Goldin, 2014), and shorter commutes (Barbanchon et al., 2021). We contribute to this literature on geographic mobility and gender inequality in labor markets by showing that mobility friction does more than drive the job segregation behind a large share of the gender wage gap (Goldin, 2014; Barbanchon et al., 2021). It continues to constrain women’s access to high-stakes professional opportunities even after they have already sorted into a job. These constraints have been studied almost entirely in the context of permanent relocation or recurring daily commuting. We extend them to episodic opportunities that require travel that is temporary, infrequent, and layered onto an existing job. Mobility frictions shape access to influential roles even when the travel they require is short-lived.

Our findings carry a practical implication for how episodic advisory work is organized. The constraint we document is a property of how the engagement is designed, not of the underlying work itself, so it is amenable to intervention. Deliberately allowing these engagements to be conducted remotely removes the constraint at its source. For Congress, retaining a virtual or hybrid testimony option would preserve the wider and more diverse and balanced pool of experts that the pandemic period briefly made accessible.

The same logic extends to any organization that convenes episodic expert advice, including outside consultants brought in for a specific engagement, independent directors attending board meetings, and technical experts convened for due diligence ahead of an investment or acquisition, to name a few. These settings often default to requiring physical presence for tasks that do not strictly need it, imposing a cost that falls unevenly across invited experts, particularly on those carrying a heavier caregiving load or less flexibility over travel. This unevenly distributed cost narrows the pool of experts willing or able to participate, so organizations capture less of the value that outside expertise could offer them. Our results suggest that what stands in the way is often not the difficulty of moving people or ideas across space, but a choice about whether moving them is required at all.

8 Limitations

Our study has several limitations. First, what we can observe is limited to the realized match, the set of academics who actually appeared before Congress, not the fuller set of candidates who were considered or invited but did not testify. This is why we construct a proxied at-risk population rather than observing the true consideration set directly. Settings with documentation of who was

invited or considered, and not only who ultimately participated, would allow future work test our mechanism against the actual choice set rather than an approximation of it.

Second, our at-risk pool is built from semantic similarity between publications. This approximates the set of experts a committee might plausibly have drawn on, but it cannot recover that set exactly. Some economists who could genuinely have been called are missed, and some we include would never have been considered. Both add noise, but neither gives us reason to expect a bias that runs specifically along the lines of gender and distance that drive our results.

Third, our design identifies selection into testimony conditional on the at-risk pool, not the underlying decision to participate. We interpret the mobility results as supply-side, women constrained from accepting geographically costly engagements, but the conditional logit cannot fully separate this from a demand-side account in which committees invite differently across gender and distance. Relatedly, we observe realized testimony rather than invitations, so we cannot say whether the rise in women’s participation under virtual formats reflects greater willingness to accept, a greater likelihood of being invited, or both. Data on invitations would let future work separate these margins directly.

Fourth, congressional testimony is one specific form of expert advisory work, and a relatively rare one at that. The mobility mechanism we document has been observed in adjacent settings, including virtual academic seminars (Biermann, 2024) and the academic job market (Bisantis, 2026), which lends it some plausibility beyond our own. Whether it extends to other forms of episodic advisory work, such as board service, consulting, or due diligence panels, and beyond economics to fields whose engagement with policy and industry is organized differently, remains a task for future work.

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Appendices

Appendix A: Prompt and Example Provided to OpenAI for the Extraction of Witnesses

```
example_json = {
  "hearings": [
    {
      "year": "2017",
      "date_begin": "2017-02-28",
      "date_end": "2017-07-18",
      "virtual_hearing": "no",
      "full_name_string": "Williams, Ph.D., Gary W.",
      "witness_name": "Gary W. Williams",
      "witness_affiliation": "Professor of Agricultural Economics and Co-Director, Agribusiness, Food, and Consumer Economics Research Center, Department of Agricultural Economics, Texas A & M University",
      "job_title": "Professor, Co-Director",
      "witness_title": "Ph.D.",
      "academic_witness": "Yes",
      "discipline": "Economics",
      "university_affiliation": "Texas A&M University",
      "full_committee": "Committee on Agriculture",
      "sub_committee": "Subcommittee on Livestock and Foreign Agriculture",
      "sub_committee_count": "6",
      "title_description": "The Next Farm Bill",
      "chamber": "House of Representatives"
    }
  ]
}

def get_messages(testo):
    return [
        {
            "role": "system", "content": "You are tasked with extracting all witness information from the following congressional hearing text. Output the witness details as JSON, with each witness being represented by a dictionary. Be thorough: witnesses might be listed consecutively or their details may be spread across lines. Typically they will appear in a consecutive list after 'WITNESSES' and be separated by a list of dots '.....'. When you think you are done, check again to ensure no witness is missed. Use this schema for each witness: "+json.dumps(example_json)}",
        },
        {
            "role": "user", "content": f"Extract the following information from the provided congressional hearing text. Each witness should be represented in this JSON structure: [{{'year': 'the year of the hearing', 'date_begin': 'the first day of the hearing', 'date_end': 'the last day of the hearing', 'virtual_hearing': 'yes when the hearing was a hybrid or online format and otherwise no.', 'full_name_string': 'the full name of the witness. Make sure you exclude the affiliation here.', 'witness_name': 'full name of the witness in the format first name middle name last name', 'witness_title': 'the title of the witness', 'academic_witness': 'yes or no', 'discipline': 'the discipline of the witness', 'university_affiliation': 'the university affiliation of the witness', 'full_committee': 'the full committee of the witness', 'sub_committee': 'the subcommittee of the witness', 'sub_committee_count': 'the number of subcommittees the witness is on', 'title_description': 'the title description of the witness', 'chamber': 'the chamber of the witness'}]"}"
        }
    ]
```

witness_affiliation': 'The full affiliation of the witness. If there is more than one affiliation, list them all.', 'job_title': 'The job title of the witness, inferred from their affiliation. If there are more, list them all separated by a comma.', 'witness_title': 'if there is a title in the name, such as honorable, PhD, Dr, or Jr., save it here. If there are more than one title, list them all.', 'academic_witness': 'yes if the witness works as an academic, otherwise no. The mention of professor and/or a university in the affiliation indicates academic status.', 'discipline': 'The witness' discipline if they are an academic, for instance, economics or psychology. If there is more than one discipline, include them all.', 'university_affiliation': 'If the witness has a university affiliation, state it here. Otherwise leave empty.', 'full_committee': 'The full committee of the hearing. If more than one full committee exists, list them all separated by a colon (:).', 'sub_committee': 'The subcommittee of the hearing.', 'sub_committee_count': 'The total number of subcommittees included in the hearing.', 'title_description': 'The title of the hearing.', 'chamber': 'House of Representatives, Senate, or Joint.'}},]. Text: {testo}"}

]

Appendix B: Additional Tables and Figures

Table B1: Top 30 Economists by Number of Congressional Appearances

Witness	Appearances	First year	Last year	Share (%)
Robert Eisner	58	1969	1997	1.09
Allan Meltzer	46	1972	2015	0.86
Martin Feldstein	44	1972	2015	0.83
Walter Heller	42	1969	1986	0.79
Jeffrey Sachs	41	1986	2010	0.77
Lester Thurow	38	1970	1999	0.71
Simon Johnson	34	2008	2016	0.64
Murray Weidenbaum	33	1972	1999	0.62
John Galbraith	31	1969	1992	0.58
John Taylor	29	1992	2023	0.55
Karen Davis	28	1981	1992	0.53
Lawrence Klein	28	1969	1989	0.53
Uwe Reinhardt	28	1975	2013	0.53
Lawrence White	27	1974	2014	0.51
Hendrik Houthakker	26	1971	1991	0.49
Alan Blinder	26	1980	2017	0.49
Gary W. Williams	24	1991	2023	0.45
Alan Auerbach	24	1981	2011	0.45
Benjamin Friedman	24	1976	2007	0.45
James Tobin	24	1971	1993	0.45
Dale Jorgenson	23	1971	2012	0.43
Rudiger Dornbusch	23	1977	1995	0.43
Richard Cooper	23	1969	2005	0.43
Alfred Kahn	22	1969	2003	0.41
Edward Kane	22	1975	2014	0.41
Paul Samuelson	21	1970	1993	0.39
Paul Roberts	20	1970	1987	0.38
Laurence Kotlikoff	19	1989	2015	0.36
Walter Adams	19	1969	1990	0.36
Francis Thomas Juster	19	1973	1994	0.36

Table B2: University Affiliation of Economists Testifying in Congress

Witness University Affiliation	# of Testimonies	Share (%)	Witness University Affiliation	# of Witnesses	Share (%)
Harvard University	512	9.63	Harvard University	112	4.84
Mit	239	4.49	University Of California Berkeley	57	2.46
University Of Pennsylvania	172	3.23	University Of Pennsylvania	55	2.38
University Of Michigan	153	2.88	George Mason University	53	2.29
Stanford University	153	2.88	University Of Maryland	52	2.25
George Mason University	134	2.52	Stanford University	51	2.20
Yale University	129	2.43	University Of Chicago	48	2.07
University Of Maryland	126	2.37	University Of Michigan	46	1.99
Princeton University	123	2.31	Mit	42	1.81
Columbia University	117	2.20	University Of Wisconsin Madison	41	1.77
University Of California Berkeley	113	2.12	Yale University	41	1.77
University Of Chicago	105	1.97	Columbia University	39	1.68
Georgetown University	101	1.90	Cornell University	36	1.56
University Of Minnesota	98	1.84	Princeton University	35	1.51
Carnegie-Mellon University	98	1.84	Georgetown University	35	1.51
Northwestern University	96	1.80	University Of Minnesota	32	1.38
Cornell University	93	1.75	Iowa State University	27	1.17
Johns Hopkins University	79	1.49	George Washington University	27	1.17
University Of Wisconsin Madison	66	1.24	Nyu	26	1.12
Michigan State University	65	1.22	Johns Hopkins University	25	1.08
George Washington University	63	1.18	Rutgers University	25	1.08
Iowa State University	62	1.17	Duke University	25	1.08
Nyu	57	1.07	University Of Illinois	23	0.99
Texas A&M University	53	1.00	Indiana University	23	0.99
University Of Texas	51	0.96	Michigan State University	23	0.99
Duke University	50	0.94	University Of California	22	0.95
Rutgers University	49	0.92	University Of Missouri	22	0.95
University Of Virginia	48	0.90	Carnegie-Mellon University	21	0.91
Ohio State University	45	0.85	University Of California Davis	20	0.86
University Of Illinois	44	0.83	Ohio State University	20	0.86

The left column is total number of testimonies per university, and the right column is number of unique economists per university.

Table B3: Committee Groupings and their Corresponding Committees

Committee Type	Committees
Agriculture	Committee on Agriculture; Committee on Agriculture, Nutrition, and Forestry; Committee on Agriculture and Forestry; Committee on Merchant Marine and Fisheries
Banking and Finance	Committee on Finance; Committee on Financial Services; Committee on Banking, Housing, and Urban Affairs; Committee on Banking, Currency, and Housing; Committee on Banking, Finance, and Urban Affairs; Committee on Banking and Financial Services; Committee on Banking and Currency; House Committee on Banking and Financial Services
Budget and Appropriations	Committee on Ways and Means; Committee on the Budget; Committee on Appropriations; Committee on Budget; Study Committee on Budget Control
Economy	Committee on Economic; Committee on Small Business and Entrepreneurship; Joint Economic Committee; Committee on Small Business; Committee on Small Business, Select; Special Committee to Study Problems of American Small Business; Economic Committee; Committee on Economic Report
Energy and Commerce	Committee on Interstate and Foreign Commerce; Committee on Energy and Commerce; Committee on Commerce; Select Committee on the Climate Crisis; Committee on Energy Independence and Global Warming, Select; Committee on Atomic Energy
Infrastructure	Committee on Public Works; Committee on Commerce, Science, and Transportation; Committee on Transportation and Infrastructure; Committee on Environment and Public Works; Committee on Public Works and Transportation; Committee on Post Office and Civil Service
Internal	Committee on Rules and Administration; Committee on House Administration; Committee on Rules; Select Committee on the Modernization of Congress; Committee on Standards and Conduct, Select; Committee on the Organization of Congress
Judiciary	Committee on the Judiciary
<i>Continued on next page</i>	

Table B3: Committee Groupings and their Corresponding Committees (continued)

Committee Type	Committees
Labor and Welfare	Committee on Children, Youth, and Families, Select; Committee on Labor and Public Welfare; Committee on Nutrition and Human Needs, Select; Committee on Health, Education, Labor, and Pensions; Special Committee on Aging; Committee on Education and the Workforce; Committee on Education and Labor; Committee on Veterans' Affairs; Committee on Labor and Human Resources; Committee on Equal Educational Opportunity, Select; Committee on Economic and Educational Opportunities; Committee on Labor-Management Relations; Joint Select Committee on Solvency of Multiemployer Pension Plans; Committee on Narcotics Abuse and Control, Select; Committee on Aging, Select; Committee on Hunger, Select; Committee on Human Resources; Committee on Crime, Select; Special Committee on Unemployment Problems
Natural Resources	Committee on Energy and Natural Resources; Committee on Natural Resources; Committee on Interior and Insular Affairs; Committee on Public Lands; Committee on Resources; Committee on National Water Resources, Select
Other	House Commissions and Temporary Committees; Committee on Indian Affairs; Committee on the District of Columbia; Commissions and Temporary Committees; Committee to Investigate the Use of Chemicals in Food Products, Select; Committee to Investigate the Use of Chemicals in Foods and Cosmetics, Select; Committee on Population, Select; Committee on Un-American Activities; Committee on Indian Affairs, Select; Committee on Democratic Policy; Committee on Washington Metropolitan Problems
Oversight	Committee on Governmental Affairs; Committee on Government Operations; Committee on Oversight and Government Reform; Committee on Oversight and Reform; Committee on Oversight and Accountability; Committee on Government Reform and Oversight; Committee on Government Reform; Committee on Oversight; House Committee on Government Reform and Oversight; Committee on Expenditures in Executive Departments
Science	Committee on Science and Technology; Committee on Science, Space, and Technology; Committee on Science; Committee on Science and Astronautics; Committee on Aeronautical and Space Sciences; Ad Hoc Select Committee on the Outer Continental Shelf

Continued on next page

Table B3: Committee Groupings and their Corresponding Committees (continued)

Committee Type	Committees
Security and Foreign Affairs	Committee on Homeland Security; Committee on Foreign Relations; Committee on Homeland Security and Governmental Affairs; Committee on Foreign Affairs; Commission on Security and Cooperation in Europe; Committee on Armed Services; Committee on International Relations; Committee on National Security; Committee to Investigate Communist Aggression and the Forced Incorporation of the Baltic States into the Soviet Union, Select; Committee on Internal Security; Select Committee on Intelligence; Congressional-Executive Commission on China; Committee on Naval Affairs

Table B4: Conditional Logit Stratified on Hearing

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female	-0.06 (0.06)	-0.06 (0.06)	-0.11* (0.06)	0.54*** (0.15)	-0.18*** (0.06)	0.58*** (0.16)	0.84*** (0.21)
log(H-Index)		0.14*** (0.03)	0.13*** (0.03)	0.13*** (0.03)	0.14*** (0.03)	0.13*** (0.03)	0.14*** (0.03)
log(Coauthors)		-0.63*** (0.02)	-0.63*** (0.02)	-0.63*** (0.02)	-0.63*** (0.02)	-0.63*** (0.02)	-0.64*** (0.02)
log(Witness Ties)		0.85*** (0.03)	0.83*** (0.03)	0.84*** (0.03)	0.83*** (0.03)	0.83*** (0.03)	0.81*** (0.05)
Academic age		0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)
log(Top 5 Publication Stock)		0.10*** (0.02)	0.08*** (0.02)	0.09*** (0.02)	0.08*** (0.02)	0.09*** (0.02)	0.09*** (0.02)
log(Policy Mentions)		-0.37*** (0.02)	-0.36*** (0.02)	-0.37*** (0.02)	-0.36*** (0.02)	-0.36*** (0.02)	-0.37*** (0.02)
log(Media Mentions)		0.13*** (0.03)	0.12*** (0.03)	0.13*** (0.03)	0.13*** (0.03)	0.13*** (0.03)	0.13*** (0.03)
Top 10 University		0.27*** (0.04)	0.36*** (0.04)	0.36*** (0.04)	0.36*** (0.04)	0.36*** (0.04)	0.37*** (0.04)
log(Driving Time)			-0.31*** (0.02)	-0.27*** (0.02)	-0.31*** (0.02)	-0.27*** (0.02)	-0.28*** (0.03)
Female x Virtual					0.84*** (0.23)	-0.09 (0.54)	
Female x log(Driving Time)				-0.34*** (0.08)		-0.40*** (0.08)	-0.62*** (0.11)
Virtual x log(Driving Time)						-0.16 (0.15)	
Female x Virtual x log(Driving Time)						0.49** (0.23)	
Female x log(Witness Ties)							-0.39** (0.17)
log(Witness Ties) x log(Driving Time)							0.00 (0.02)
Female x log(Witness Ties) x log(Driving Time)							0.36*** (0.08)
Obs.	52397	52397	52397	52397	52397	52397	52397
Testimonies	3662	3662	3662	3662	3662	3662	3662
Clusters	2817	2817	2817	2817	2817	2817	2817
Log-likelihood	-9651.25	-8873.05	-8766.80	-8754.07	-8759.76	-8744.33	-8732.24

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Coefficients are reported as log-odds. Standard errors in parentheses are clustered at the hearing level. The dependent variable is an indicator for whether the economist appeared as an expert witness in a given hearing.

Estimation by conditional logit models. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

SMALL BUSINESS FRANCHISING: AN OVERVIEW OF THE INDUSTRY, SBA'S ROLE, AND LEGISLATIVE PROPOSALS

WEDNESDAY, MARCH 16, 2022

UNITED STATES SENATE,
COMMITTEE ON SMALL BUSINESS
AND ENTREPRENEURSHIP,
Washington, DC.

The Committee met, pursuant to notice, at 2:32 p.m., via WebEx in Room 301, Russell Senate Office Building, Hon. Benjamin L. Cardin, Chairman of the Committee, presiding.
Present: Senators Cardin, Cantwell, Booker, Rosen, Hickenlooper, Paul, Scott, Ernst, Young, Hawley, and Marshall.

Figure B1: Cover Page of a Virtual Senate Hearing Transcript

Notes: Cover page of the transcript for the U.S. Senate Committee on Small Business and Entrepreneurship hearing “Small Business Franchising: An Overview of the Industry, SBA’s Role, and Legislative Proposals,” held March 16, 2022. The transcript records that the Committee met via WebEx, with Senator Benjamin L. Cardin presiding as Chairman.

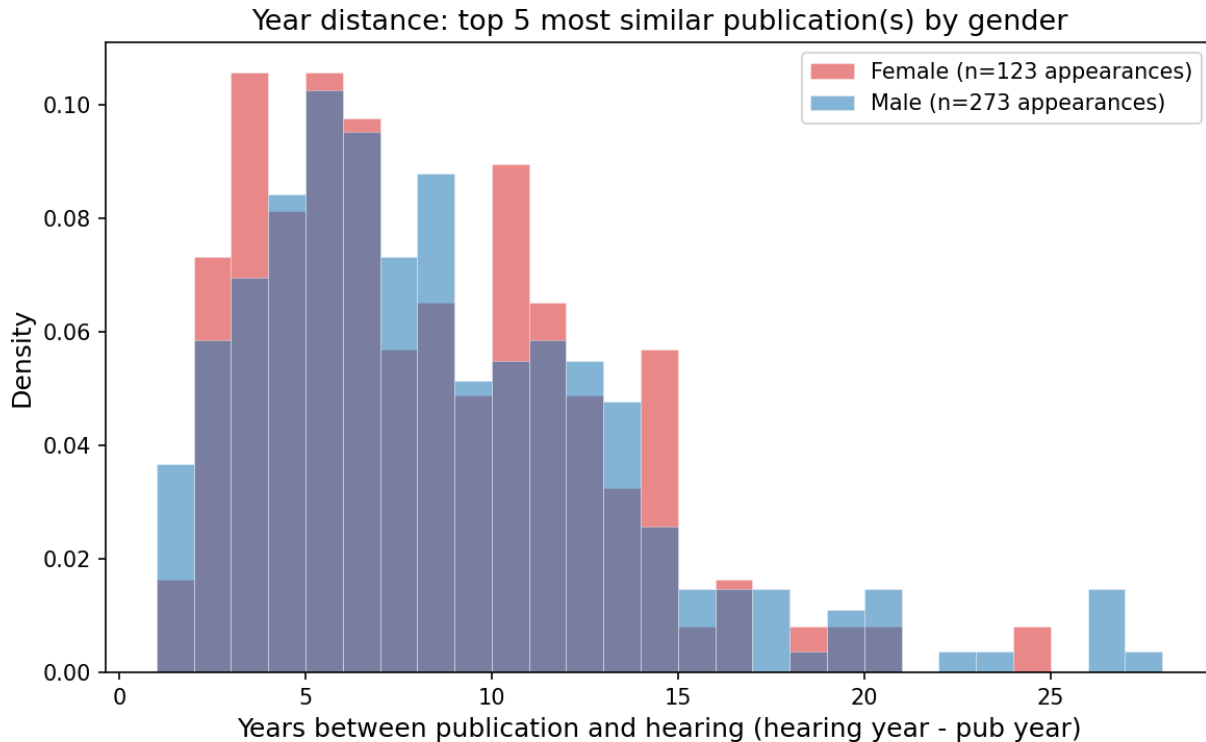


Figure B2: Years Between most Similar Publications and Hearing

Notes: For all hearings between 2017-2023, the witness' testimony has been extracted from the transcript, encoded using LogicMill, and the cosine similarity has been calculated against the witness' stock of publications. For each witness, the time between publication and testimony for the top-5 most similar publications are plotted here.

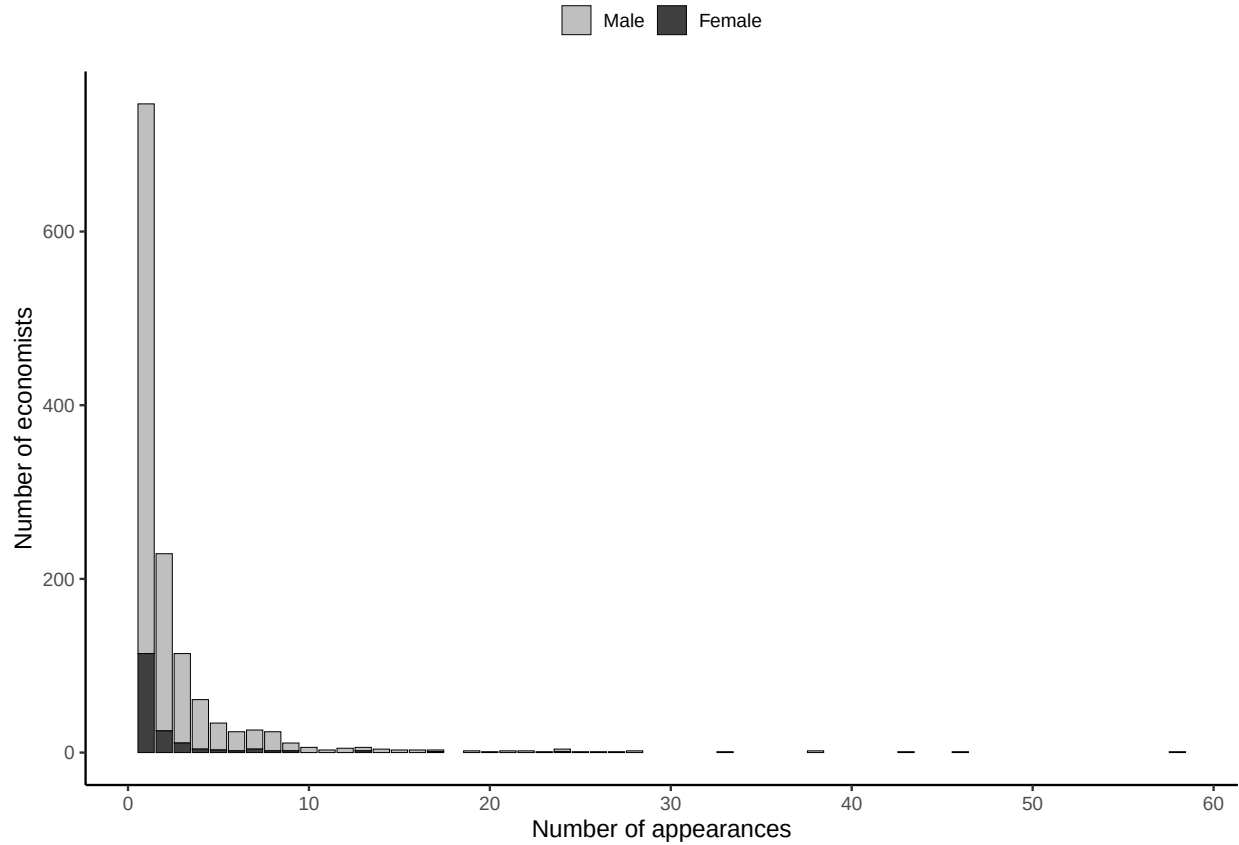


Figure B3: Frequency of Witnesses' Congressional Appearances, by Gender

Notes: Each bar shows the number of unique economists who testified a given number of times, split by gender. An economist's appearance count is the number of distinct hearings at which they testified across the full sample period.

Appendix C: Robustness Checks

Period Covered

Table C1: Restricting to Hearings from 2000 Onward

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female	0.10 (0.08)	0.11 (0.09)	0.09 (0.09)	0.49*** (0.18)	-0.03 (0.10)	0.53*** (0.20)	0.40 (0.28)
log(H-Index)		-0.34*** (0.07)	-0.33*** (0.07)	-0.33*** (0.07)	-0.33*** (0.07)	-0.33*** (0.07)	-0.34*** (0.07)
log(Coauthors)		-0.54*** (0.05)	-0.55*** (0.05)	-0.55*** (0.05)	-0.55*** (0.05)	-0.55*** (0.05)	-0.56*** (0.05)
log(Witness Ties)		0.84*** (0.05)	0.79*** (0.05)	0.80*** (0.05)	0.79*** (0.05)	0.79*** (0.05)	0.64*** (0.08)
Academic age		0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)
log(Top 5 Publication Stock)		0.08 (0.05)	0.10** (0.05)	0.11** (0.05)	0.10* (0.05)	0.10** (0.05)	0.11** (0.05)
log(Policy Mentions)		-0.22*** (0.03)	-0.22*** (0.03)	-0.23*** (0.03)	-0.22*** (0.03)	-0.22*** (0.03)	-0.23*** (0.03)
log(Media Mentions)		0.14*** (0.04)	0.13*** (0.04)	0.13*** (0.04)	0.13*** (0.04)	0.13*** (0.04)	0.13*** (0.04)
Top 10 University		0.32*** (0.09)	0.53*** (0.09)	0.53*** (0.09)	0.53*** (0.09)	0.53*** (0.09)	0.55*** (0.09)
log(Driving Time)			-0.47*** (0.04)	-0.42*** (0.04)	-0.47*** (0.04)	-0.43*** (0.05)	-0.50*** (0.07)
Female x Virtual					0.69*** (0.25)	-0.08 (0.55)	
Female x log(Driving Time)				-0.22** (0.09)		-0.31*** (0.10)	-0.36** (0.14)
Virtual x log(Driving Time)						-0.02 (0.15)	
Female x Virtual x log(Driving Time)						0.41* (0.24)	
Female x log(Witness Ties)							0.01 (0.19)
log(Witness Ties) x log(Driving Time)							0.06 (0.04)
Female x log(Witness Ties) x log(Driving Time)							0.16* (0.09)
Obs.	14301	14301	14301	14301	14301	14301	14301
Testimonies	1100	1100	1100	1100	1100	1100	1100
Clusters	942	942	942	942	942	942	942
Log-likelihood	-2786.04	-2524.56	-2439.45	-2436.54	-2435.08	-2430.07	-2425.03

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

The sample is restricted to hearings from 2000. All models are estimated by conditional logit within a matched set, where each matched set consists of the witness and the at-risk economists associated with a given hearing.

Coefficients are reported as log-odds. Standard errors in parentheses are clustered at the hearing level. The dependent variable is an indicator for whether the economist appeared as an expert witness in a given hearing.

Estimation by conditional logit models. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table C2: Restricting to Hearings from 2010 Onward

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female	0.18*	0.15	0.14	0.21	-0.02	0.20	0.21
	(0.10)	(0.11)	(0.11)	(0.22)	(0.12)	(0.24)	(0.34)
log(H-Index)		-0.39***	-0.39***	-0.39***	-0.38***	-0.38***	-0.39***
		(0.09)	(0.10)	(0.10)	(0.10)	(0.09)	(0.10)
log(Coauthors)		-0.53***	-0.53***	-0.53***	-0.54***	-0.54***	-0.54***
		(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)
log(Witness Ties)		0.85***	0.81***	0.81***	0.81***	0.81***	0.63***
		(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.11)
Academic age		0.01	0.01	0.01	0.00	0.00	0.01
		(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
log(Top 5 Publication Stock)		0.01	0.05	0.05	0.04	0.05	0.05
		(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)
log(Policy Mentions)		-0.16***	-0.17***	-0.17***	-0.17***	-0.17***	-0.17***
		(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
log(Media Mentions)		0.14***	0.13***	0.13***	0.13***	0.13***	0.13***
		(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
Top 10 University		0.39***	0.61***	0.61***	0.60***	0.61***	0.63***
		(0.11)	(0.11)	(0.11)	(0.11)	(0.11)	(0.12)
log(Driving Time)			-0.50***	-0.49***	-0.51***	-0.50***	-0.58***
			(0.05)	(0.06)	(0.05)	(0.06)	(0.08)
Female x Virtual					0.65**	0.17	
					(0.26)	(0.57)	
Female x log(Driving Time)				-0.04		-0.12	-0.21
				(0.10)		(0.12)	(0.16)
Virtual x log(Driving Time)						0.04	
						(0.15)	
Female x Virtual x log(Driving Time)						0.24	
						(0.25)	
Female x log(Witness Ties)							-0.06
							(0.22)
log(Witness Ties) x log(Driving Time)							0.07
							(0.05)
Female x log(Witness Ties) x log(Driving Time)							0.18*
							(0.10)
Obs.	8380	8380	8380	8380	8380	8380	8380
Testimonies	674	674	674	674	674	674	674
Clusters	584	584	584	584	584	584	584
Log-likelihood	-1673.74	-1510.88	-1449.43	-1449.35	-1445.97	-1444.88	-1440.08

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

The sample is restricted to hearings from 2010. All models are estimated by conditional logit within a matched set, where each matched set consists of the witness and the at-risk economists associated with a given hearing.

Coefficients are reported as log-odds. Standard errors in parentheses are clustered at the hearing level. The dependent variable is an indicator for whether the economist appeared as an expert witness in a given hearing.

Estimation by conditional logit models. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

At-Risk Pool Size

Table C3: Restricting the At-Risk Pool to the Top Five Candidates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female	-0.05 (0.06)	-0.18*** (0.07)	-0.24*** (0.07)	0.23 (0.16)	-0.32*** (0.07)	0.25 (0.17)	0.43** (0.22)
log(H-Index)		-0.22*** (0.04)	-0.24*** (0.04)	-0.24*** (0.04)	-0.24*** (0.04)	-0.23*** (0.04)	-0.23*** (0.04)
log(Coauthors)		-0.67*** (0.03)	-0.66*** (0.03)	-0.66*** (0.03)	-0.66*** (0.03)	-0.66*** (0.03)	-0.66*** (0.03)
log(Witness Ties)		0.79** (0.03)	0.77** (0.03)	0.77** (0.03)	0.77** (0.03)	0.77** (0.03)	0.80*** (0.06)
Academic age		0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)
log(Top 5 Publication Stock)		0.16*** (0.03)	0.14*** (0.03)	0.15*** (0.03)	0.14*** (0.03)	0.14*** (0.03)	0.15*** (0.03)
log(Policy Mentions)		-0.35*** (0.03)	-0.34*** (0.03)	-0.34*** (0.03)	-0.33*** (0.03)	-0.34*** (0.03)	-0.34*** (0.03)
log(Media Mentions)		0.16*** (0.04)	0.14*** (0.04)	0.15*** (0.04)	0.16*** (0.04)	0.16*** (0.04)	0.13*** (0.04)
Top 10 University		0.33*** (0.04)	0.41*** (0.05)	0.41*** (0.05)	0.41*** (0.05)	0.41*** (0.05)	0.43*** (0.05)
log(Driving Time)			-0.29*** (0.02)	-0.27*** (0.02)	-0.30*** (0.02)	-0.27*** (0.02)	-0.25*** (0.03)
Female x Virtual					1.02*** (0.26)	0.36 (0.50)	
Female x log(Driving Time)				-0.25*** (0.08)		-0.30*** (0.09)	-0.49*** (0.11)
Virtual x log(Driving Time)						0.03 (0.13)	
Female x Virtual x log(Driving Time)						0.34* (0.21)	
Female x log(Witness Ties)							-0.30 (0.19)
log(Witness Ties) x log(Driving Time)							-0.03 (0.03)
Female x log(Witness Ties) x log(Driving Time)							0.33*** (0.09)
Obs.	21972	21972	21972	21972	21972	21972	21972
Testimonies	3662	3662	3662	3662	3662	3662	3662
Clusters	2817	2817	2817	2817	2817	2817	2817
Log-likelihood	-6561.12	-5790.02	-5707.59	-5702.07	-5699.14	-5691.88	-5683.19

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

The sample is restricted to the top five at-risk candidates per witness, whereas the main sample includes up to 20.

At-risk candidates are ranked by the number of published articles that are matched to the witness, and the semantic similarity scores of those matched articles. All models are estimated by conditional logit within a matched set, where each matched set consists of the witness and the at-risk economists associated with a given hearing.

Coefficients are reported as log-odds. Standard errors in parentheses are clustered at the hearing level. The dependent variable is an indicator for whether the economist appeared as an expert witness in a given hearing.

Estimation by conditional logit models. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table C4: Restricting the At-Risk Pool to the Top Ten Candidates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female	-0.06 (0.06)	-0.07 (0.06)	-0.11* (0.06)	0.52*** (0.15)	-0.18*** (0.07)	0.55*** (0.16)	0.80*** (0.21)
log(H-Index)		0.03 (0.03)	0.02 (0.03)	0.02 (0.03)	0.02 (0.03)	0.02 (0.03)	0.03 (0.03)
log(Coauthors)		-0.64*** (0.02)	-0.63*** (0.02)	-0.63*** (0.02)	-0.63*** (0.02)	-0.64*** (0.02)	-0.64*** (0.02)
log(Witness Ties)		0.84*** (0.03)	0.82*** (0.03)	0.82*** (0.03)	0.82*** (0.03)	0.82*** (0.03)	0.80*** (0.06)
Academic age		0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)
log(Top 5 Publication Stock)		0.12*** (0.02)	0.10*** (0.02)	0.10*** (0.02)	0.10*** (0.02)	0.10*** (0.02)	0.11*** (0.02)
log(Policy Mentions)		-0.36*** (0.02)	-0.36*** (0.02)	-0.36*** (0.02)	-0.35*** (0.02)	-0.36*** (0.02)	-0.36*** (0.02)
log(Media Mentions)		0.15*** (0.03)	0.14*** (0.03)	0.14*** (0.03)	0.14*** (0.03)	0.15*** (0.04)	0.14*** (0.03)
Top 10 University		0.29*** (0.04)	0.38*** (0.04)	0.38*** (0.04)	0.38*** (0.04)	0.38*** (0.04)	0.39*** (0.04)
log(Driving Time)			-0.30*** (0.02)	-0.26*** (0.02)	-0.30*** (0.02)	-0.26*** (0.02)	-0.26*** (0.03)
Female x Virtual					0.87*** (0.24)	-0.00 (0.52)	
Female x log(Driving Time)				-0.33*** (0.08)		-0.39*** (0.08)	-0.61*** (0.11)
Virtual x log(Driving Time)						-0.12 (0.14)	
Female x Virtual x log(Driving Time)						0.47** (0.22)	
Female x log(Witness Ties)							-0.40** (0.18)
log(Witness Ties) x log(Driving Time)							-0.00 (0.02)
Female x log(Witness Ties) x log(Driving Time)							0.37*** (0.08)
Obs.	39315	39315	39315	39315	39315	39315	39315
Testimonies	3662	3662	3662	3662	3662	3662	3662
Clusters	2817	2817	2817	2817	2817	2817	2817
Log-likelihood	-8677.16	-7917.78	-7819.85	-7808.27	-7812.49	-7798.50	-7787.41

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

The sample is restricted to the top ten at-risk candidates per witness, whereas the main sample includes up to 20.

At-risk candidates are ranked by the number of published articles that are matched to the witness, and the semantic similarity scores of those matched articles. All models are estimated by conditional logit within a matched set, where each matched set consists of the witness and the at-risk economists associated with a given hearing.

Coefficients are reported as log-odds. Standard errors in parentheses are clustered at the hearing level. The dependent variable is an indicator for whether the economist appeared as an expert witness in a given hearing.

Estimation by conditional logit models. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table C5: Restricting the At-Risk Pool to the Top Fifteen Candidates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female	-0.06 (0.06)	-0.06 (0.06)	-0.11* (0.06)	0.54*** (0.15)	-0.18*** (0.06)	0.58*** (0.16)	0.83*** (0.21)
log(H-Index)		0.13*** (0.03)	0.12*** (0.03)	0.12*** (0.03)	0.12*** (0.03)	0.12*** (0.03)	0.12*** (0.03)
log(Coauthors)		-0.63*** (0.02)	-0.63*** (0.02)	-0.63*** (0.02)	-0.63*** (0.02)	-0.63*** (0.02)	-0.64*** (0.02)
log(Witness Ties)		0.85*** (0.03)	0.84*** (0.03)	0.84*** (0.03)	0.84*** (0.03)	0.84*** (0.03)	0.81*** (0.05)
Academic age		0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)	0.01*** (0.00)
log(Top 5 Publication Stock)		0.10*** (0.02)	0.09*** (0.02)	0.09*** (0.02)	0.08*** (0.02)	0.09*** (0.02)	0.09*** (0.02)
log(Policy Mentions)		-0.37*** (0.02)	-0.36*** (0.02)	-0.37*** (0.02)	-0.36*** (0.02)	-0.36*** (0.02)	-0.37*** (0.02)
log(Media Mentions)		0.13*** (0.03)	0.13*** (0.03)	0.13*** (0.03)	0.13*** (0.03)	0.13*** (0.03)	0.13*** (0.03)
Top 10 University		0.28*** (0.04)	0.37*** (0.04)	0.37*** (0.04)	0.37*** (0.04)	0.37*** (0.04)	0.38*** (0.04)
log(Driving Time)			-0.31*** (0.02)	-0.27*** (0.02)	-0.31*** (0.02)	-0.27*** (0.02)	-0.28*** (0.03)
Female x Virtual					0.84*** (0.23)	-0.07 (0.53)	
Female x log(Driving Time)				-0.35*** (0.08)		-0.40*** (0.08)	-0.62*** (0.11)
Virtual x log(Driving Time)						-0.15 (0.14)	
Female x Virtual x log(Driving Time)						0.48** (0.23)	
Female x log(Witness Ties)							-0.38** (0.17)
log(Witness Ties) x log(Driving Time)							0.00 (0.02)
Female x log(Witness Ties) x log(Driving Time)							0.35*** (0.08)
Obs.	50643	50643	50643	50643	50643	50643	50643
Testimonies	3662	3662	3662	3662	3662	3662	3662
Clusters	2817	2817	2817	2817	2817	2817	2817
Log-likelihood	-9549.19	-8768.21	-8661.04	-8648.26	-8654.04	-8638.69	-8626.79

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

The sample is restricted to the top fifteen at-risk candidates per witness, whereas the main sample includes up to 20. At-risk candidates are ranked by the number of published articles that are matched to the witness, and the semantic similarity scores of those matched articles. All models are estimated by conditional logit within a matched set, where each matched set consists of the witness and the at-risk economists associated with a given hearing.

Coefficients are reported as log-odds. Standard errors in parentheses are clustered at the hearing level. The dependent variable is an indicator for whether the economist appeared as an expert witness in a given hearing.

Estimation by conditional logit models. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Other: Including Outliers

Table C6: Including At-Risk Outliers

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female	-0.05 (0.06)	-0.08 (0.06)	-0.12** (0.06)	0.43*** (0.13)	-0.19*** (0.06)	0.43*** (0.13)	0.46** (0.18)
log(H-Index)		0.11*** (0.03)	0.11*** (0.03)	0.11*** (0.03)	0.11*** (0.03)	0.11*** (0.03)	0.11*** (0.03)
log(Coauthors)		-0.62*** (0.02)	-0.63*** (0.02)	-0.63*** (0.02)	-0.63*** (0.02)	-0.63*** (0.02)	-0.63*** (0.02)
log(Witness Ties)		0.85*** (0.03)	0.83*** (0.03)	0.84*** (0.03)	0.83*** (0.03)	0.84*** (0.03)	0.70*** (0.05)
Academic age		0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)
log(Top 5 Publication Stock)		0.13*** (0.02)	0.12*** (0.02)	0.12*** (0.02)	0.12*** (0.02)	0.12*** (0.02)	0.12*** (0.02)
log(Policy Mentions)		-0.31*** (0.02)	-0.31*** (0.02)	-0.31*** (0.02)	-0.30*** (0.02)	-0.31*** (0.02)	-0.31*** (0.02)
log(Media Mentions)		0.12*** (0.03)	0.12*** (0.03)	0.12*** (0.03)	0.12*** (0.03)	0.13*** (0.03)	0.12*** (0.03)
Top 10 University		0.58*** (0.04)	0.65*** (0.04)	0.65*** (0.04)	0.65*** (0.04)	0.65*** (0.04)	0.65*** (0.04)
log(Driving Time)			-0.21*** (0.02)	-0.18*** (0.02)	-0.22*** (0.02)	-0.18*** (0.02)	-0.23*** (0.03)
Female x Virtual					0.93*** (0.21)	0.14 (0.45)	
Female x log(Driving Time)				-0.20*** (0.06)		-0.33*** (0.07)	-0.46*** (0.09)
Virtual x log(Driving Time)						-0.15 (0.12)	
Female x Virtual x log(Driving Time)						0.42** (0.19)	
Female x log(Witness Ties)							-0.06 (0.15)
log(Witness Ties) x log(Driving Time)							0.06*** (0.02)
Female x log(Witness Ties) x log(Driving Time)							0.23*** (0.07)
Obs.	76904	76904	76904	76904	76904	76904	76904
Testimonies	3986	3986	3986	3986	3986	3986	3986
Clusters	3120	3120	3120	3120	3120	3120	3120
Log-likelihood	-11622.42	-10541.74	-10478.41	-10466.79	-10469.13	-10455.19	-10436.83

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

The sample includes the outliers that are removed from the main sample. The main sample excludes at-risk candidates without a US or academic affiliation, with academic seniority above 65 years, or with more than 1,144 cumulative coauthors (the 99th percentile of the distribution). Witnesses are unaffected by these restrictions in either sample. All models are estimated by conditional logit within a matched set, where each matched set consists of the witness and the at-risk economists associated with a given hearing. Coefficients are reported as log-odds. Standard errors in parentheses are clustered at the hearing level. The dependent variable is an indicator for whether the economist appeared as an expert witness in a given hearing. Estimation by conditional logit models. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.